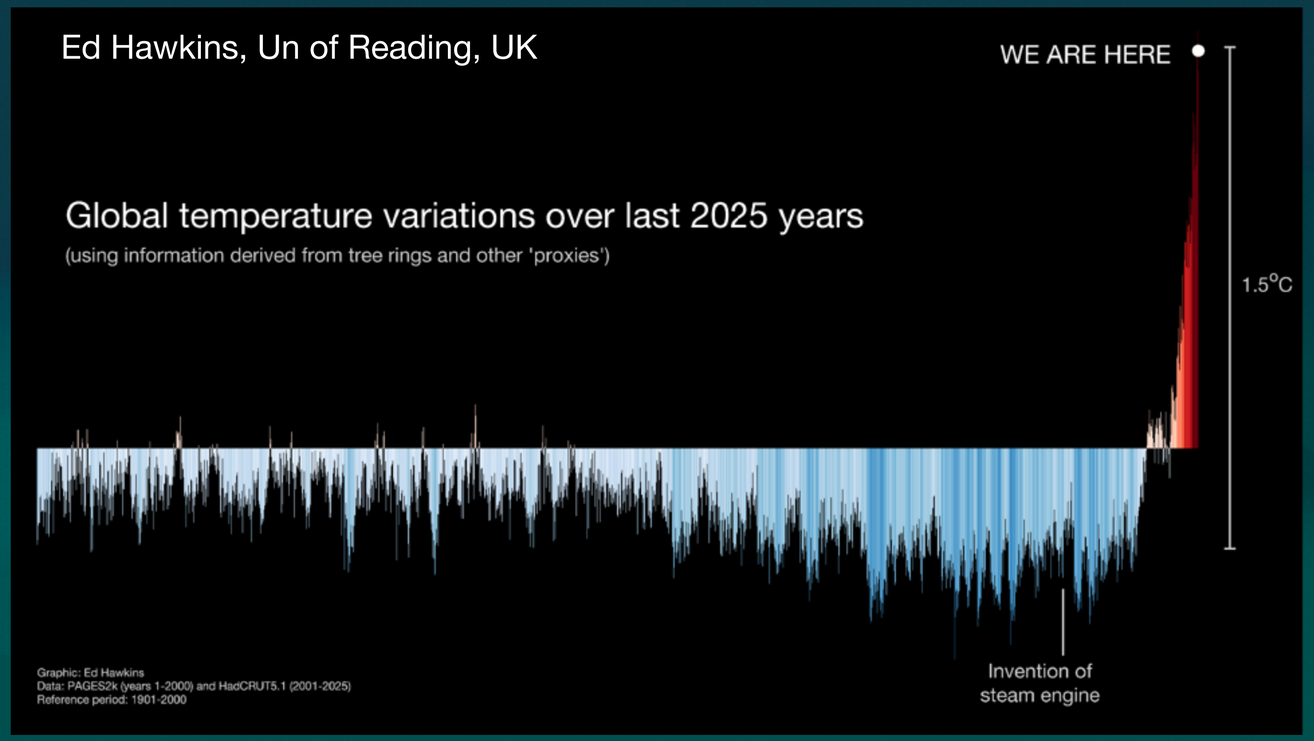
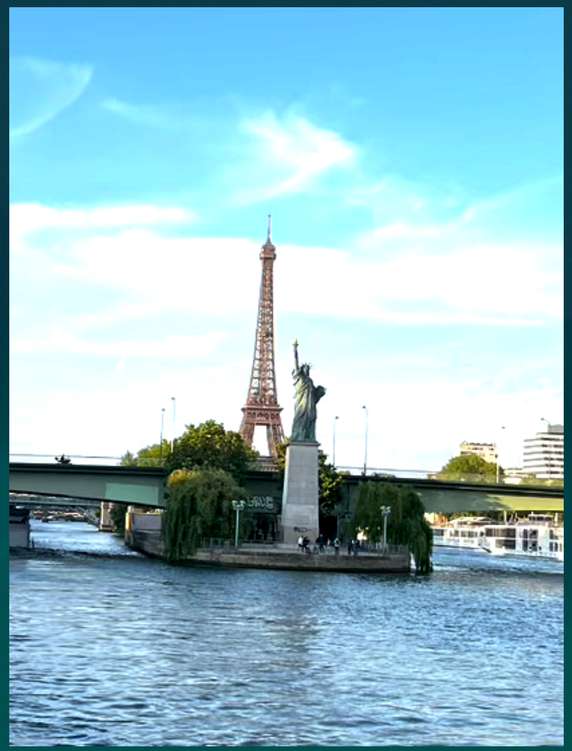
The Ocean at the Heart of a Warming Earth Climate

Stephen Griffies
IPSL/LOCEAN/CNRS + Princeton University
Seminar given at IPSL, Paris
20 October 2026

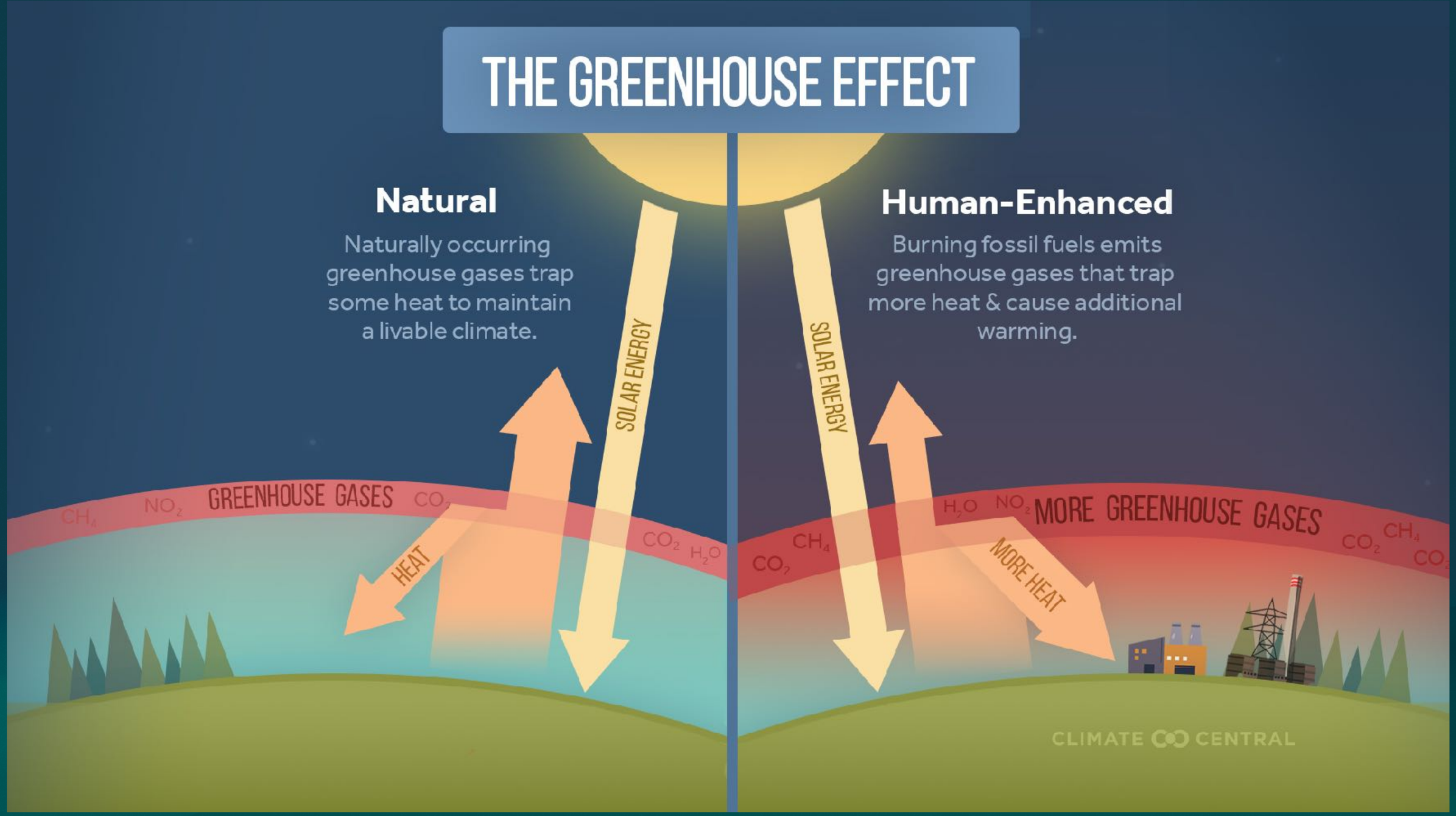




Aim and Audience

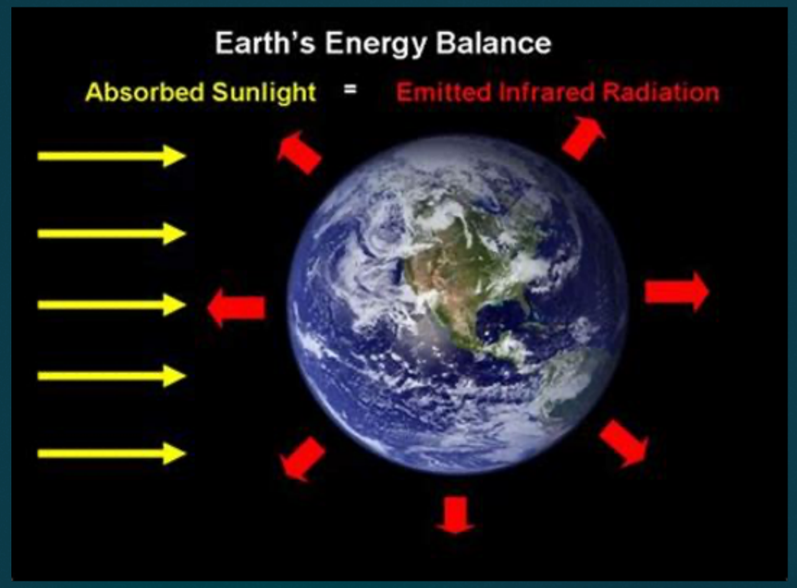
- We consider some facts and concepts about the ocean and its role in climate.
- We set the stage for ocean and climate science questions, with more to come in talks from other speakers in the seminar series.
- The presentation is aimed at non-experts interested in the science of ocean/climate physics (though hopefully of interest to experts as well).

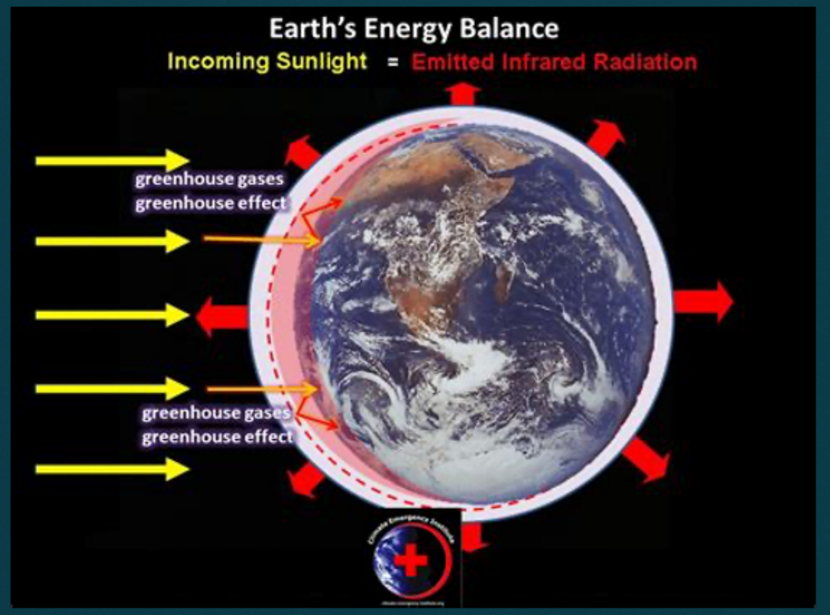
A Changing Climate

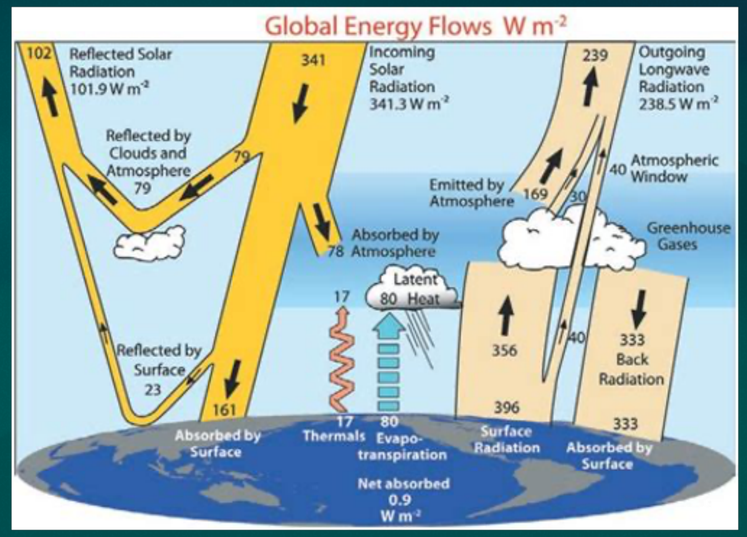


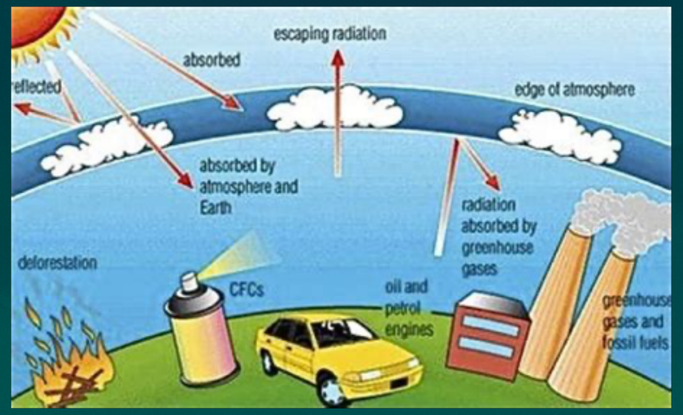
Infographic from Climate Central (Princeton USA)

Earth Energy Balance/Imbalance









Changes over two millennia

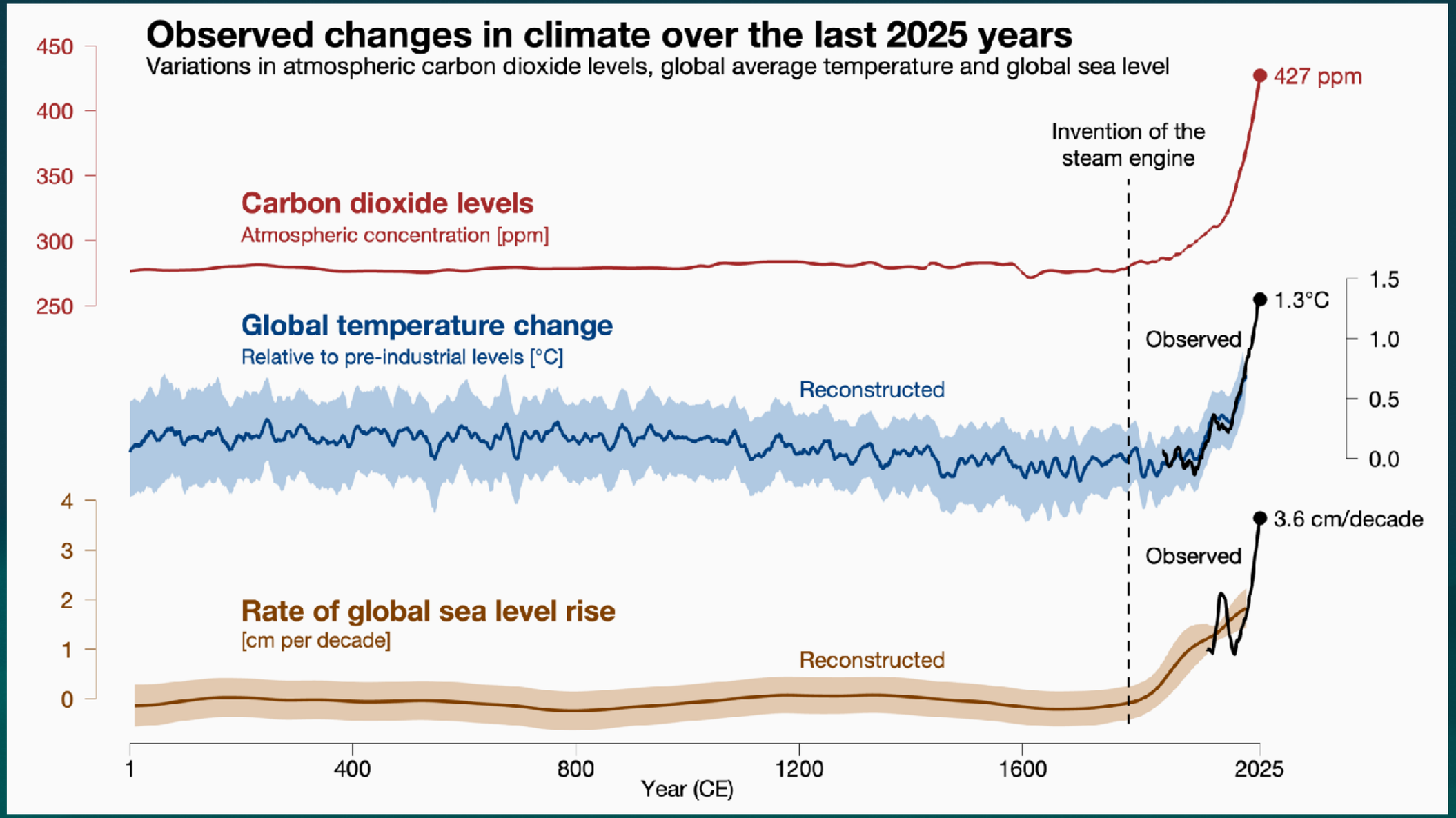
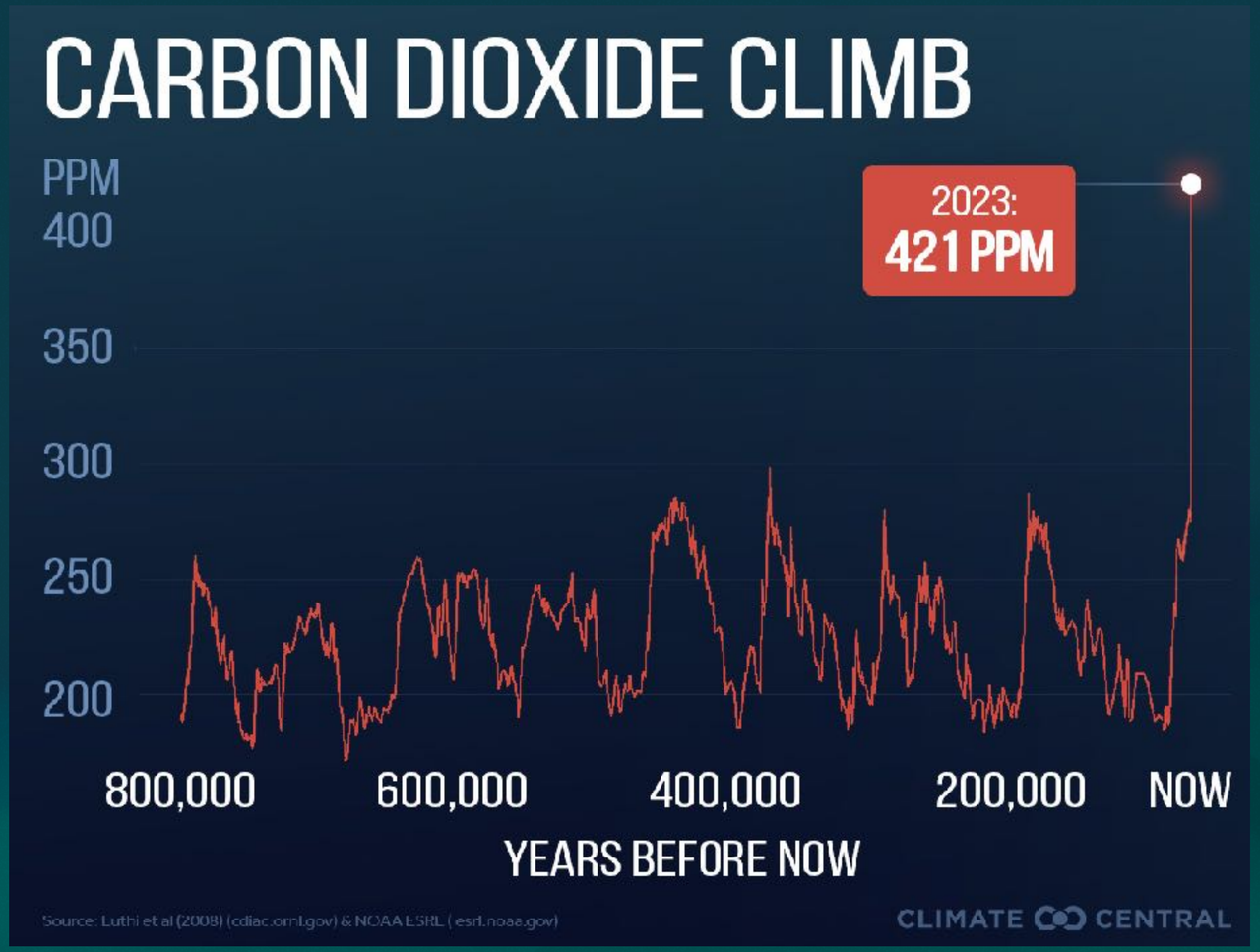
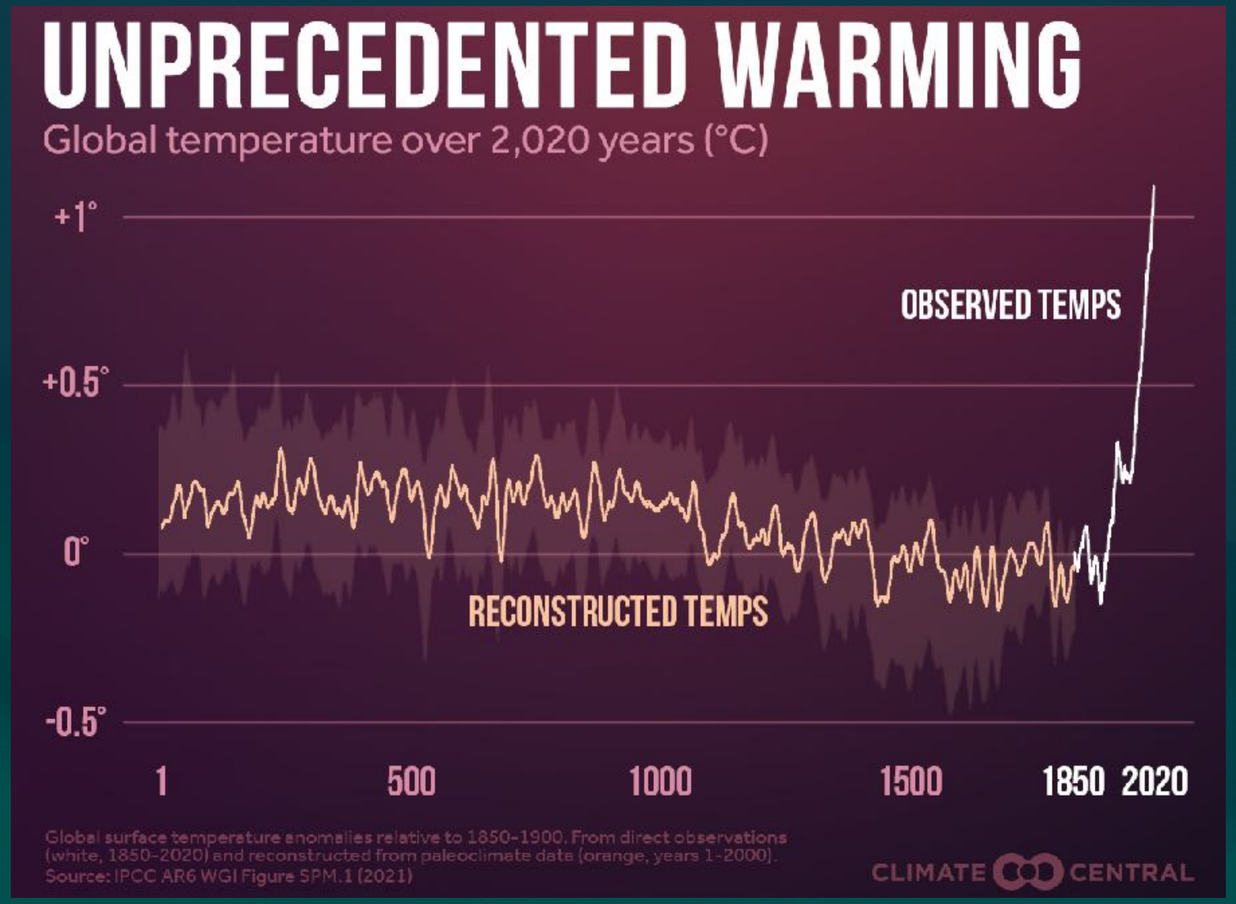


Image from Prof. Ed Hawkins, Un of Reading, UK

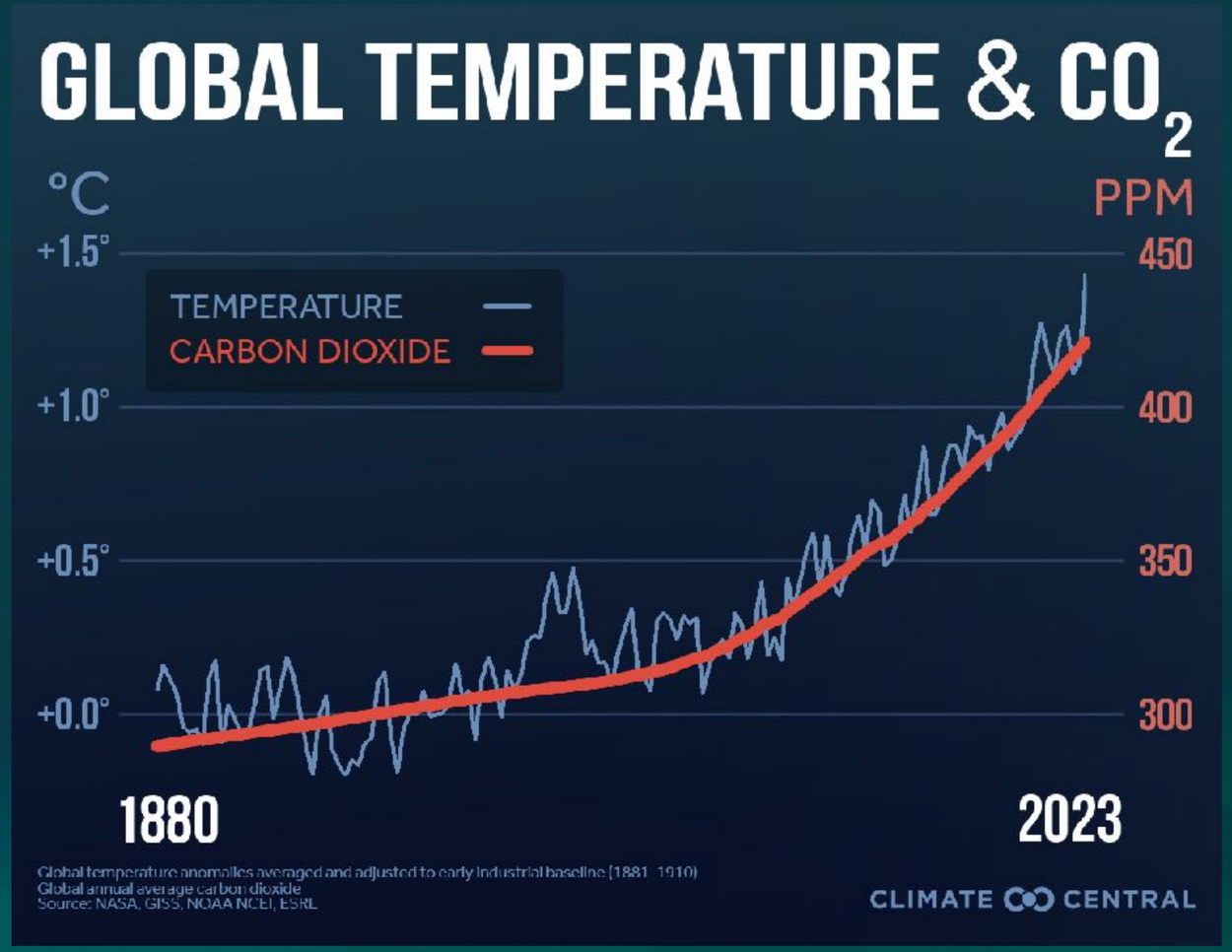
Climate and CO2 Over Time

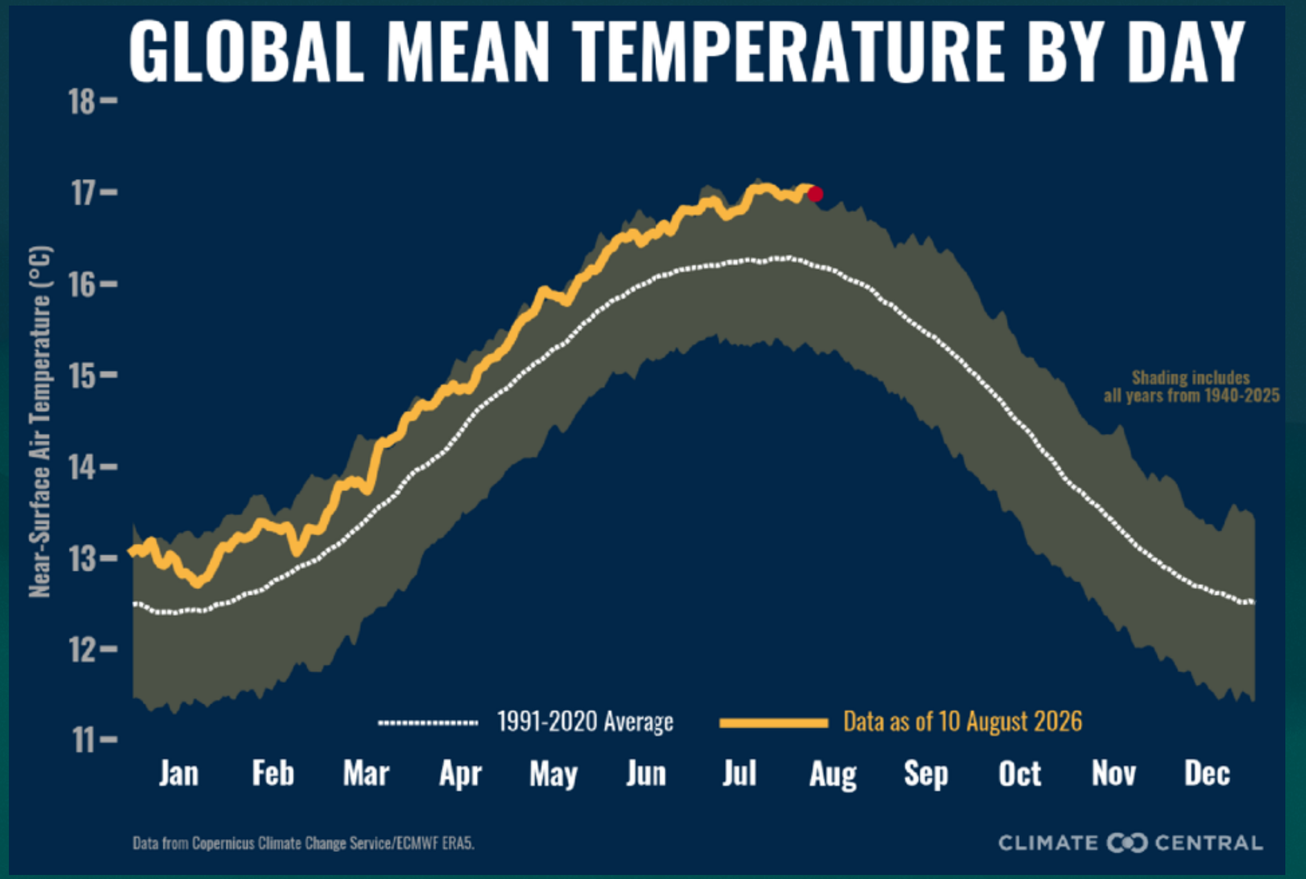




Graphics from Climate Central (Princeton USA)

Climate and CO2 During the Industrial Age

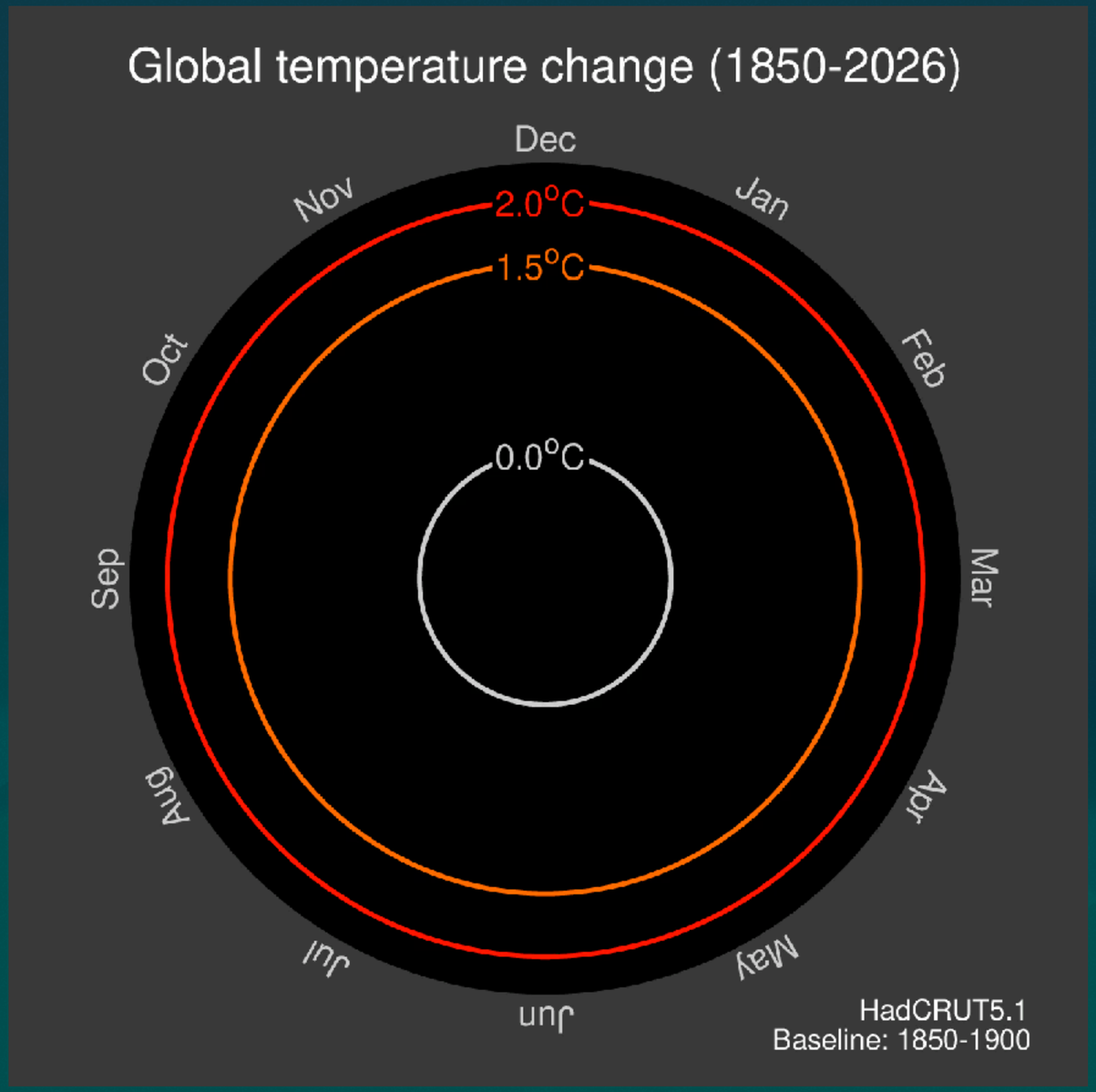




Graphics from Climate Central (Princeton USA)

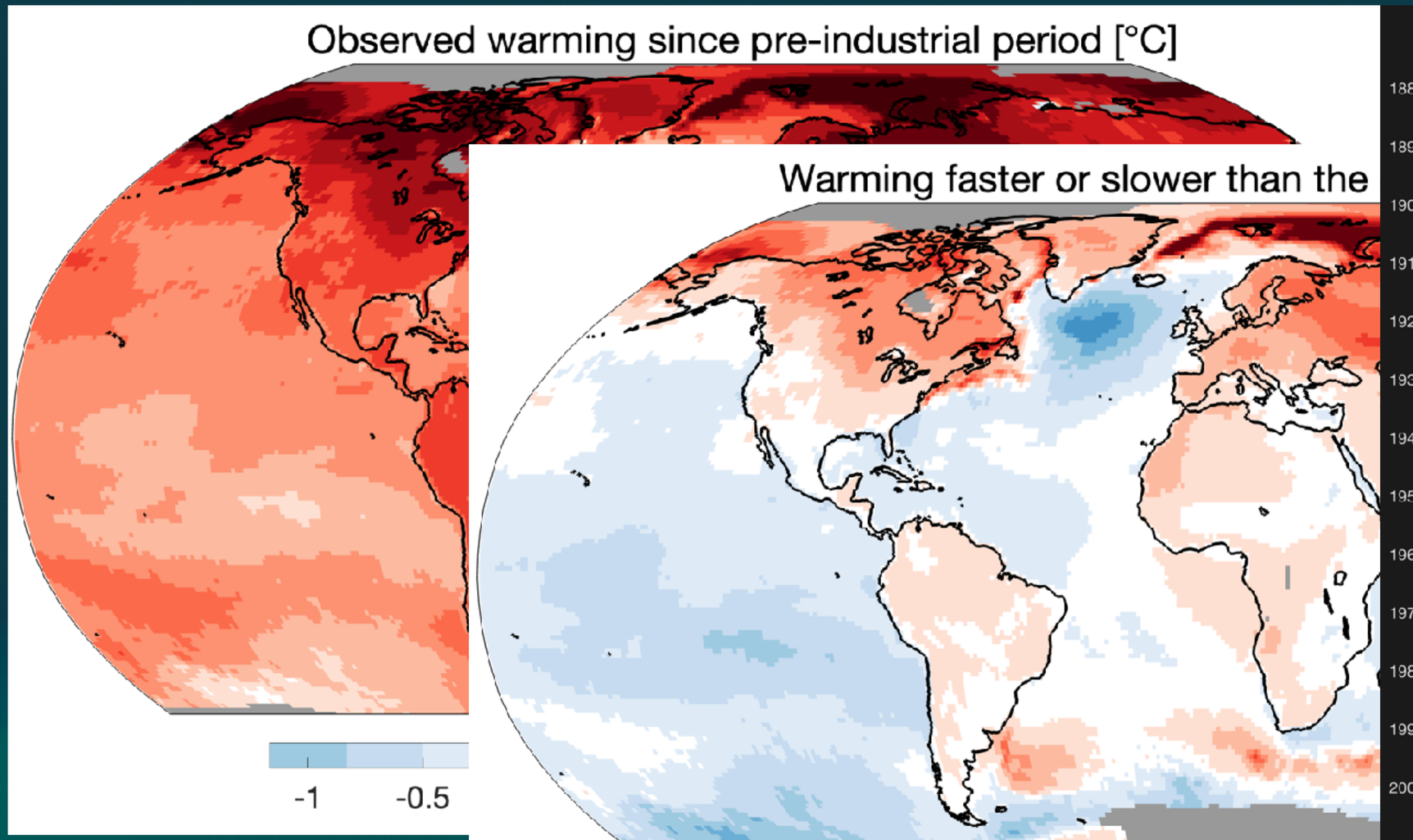
Climate Spiral

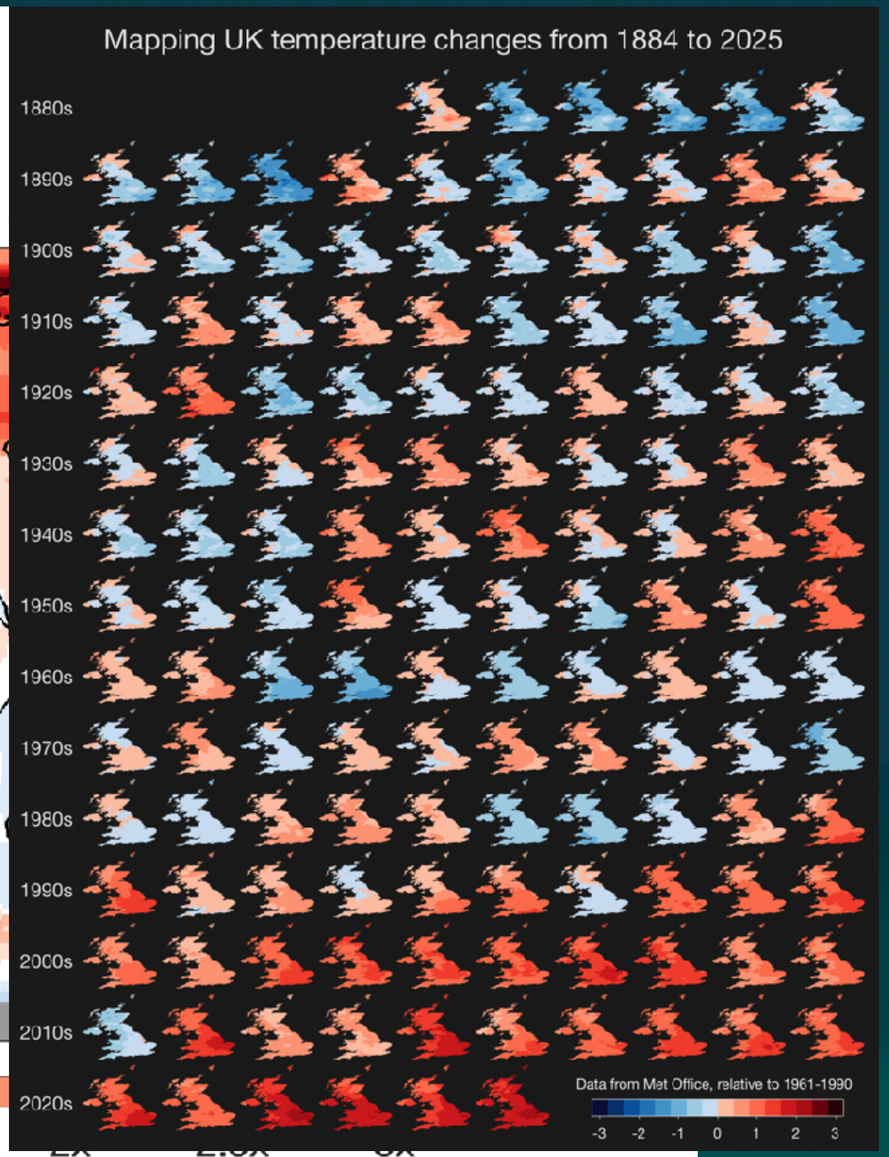
Images from Prof. Ed Hawkins, Un of Reading, UK



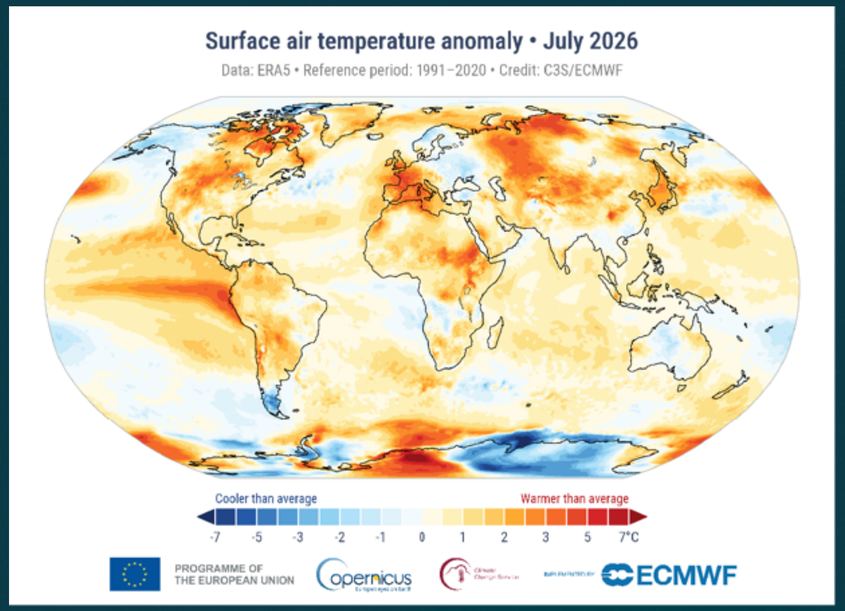
Maps of Climate Warming

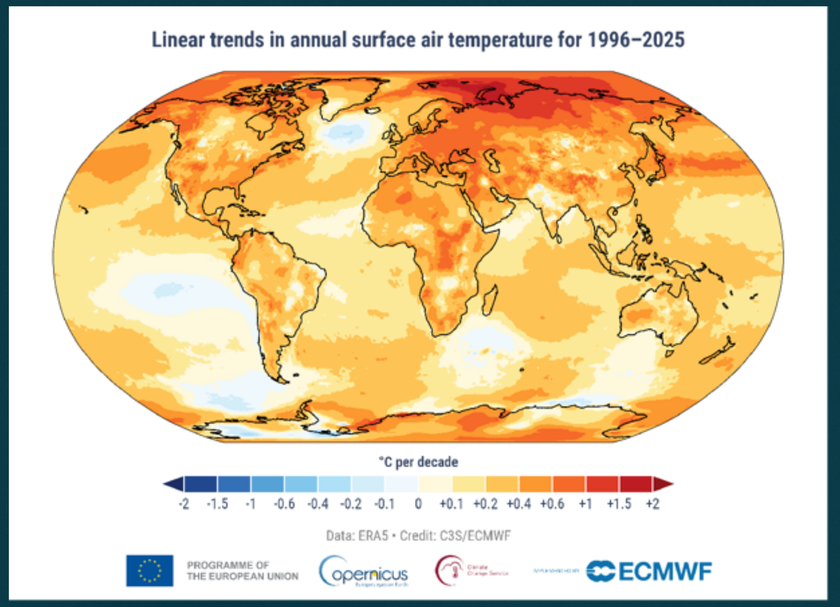
Images from Prof. Ed Hawkins, Un of Reading, UK

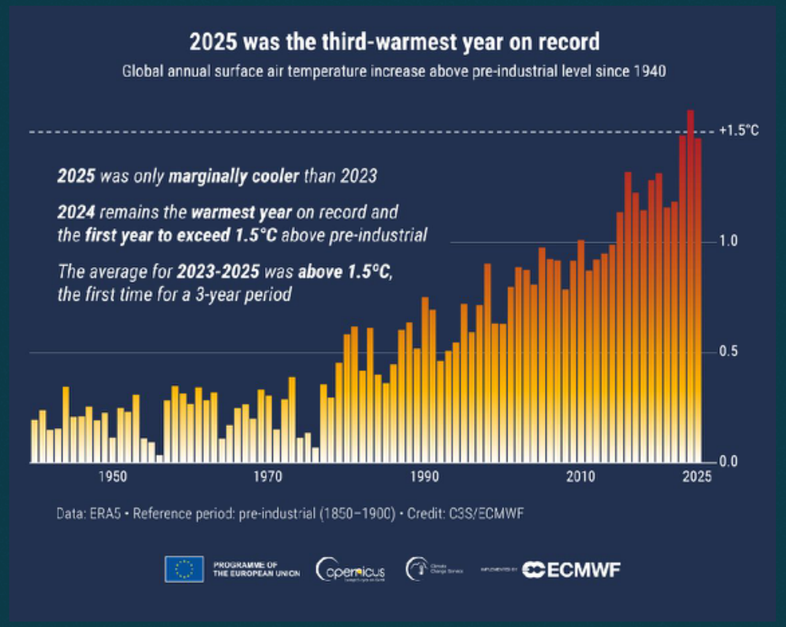


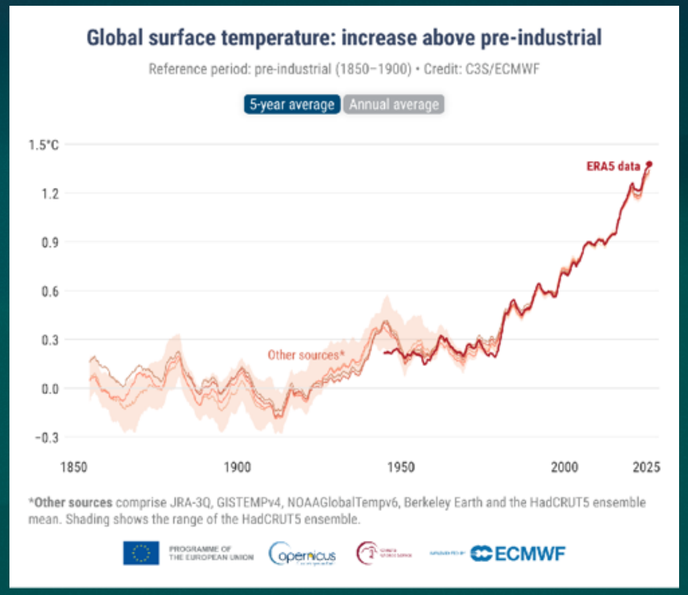


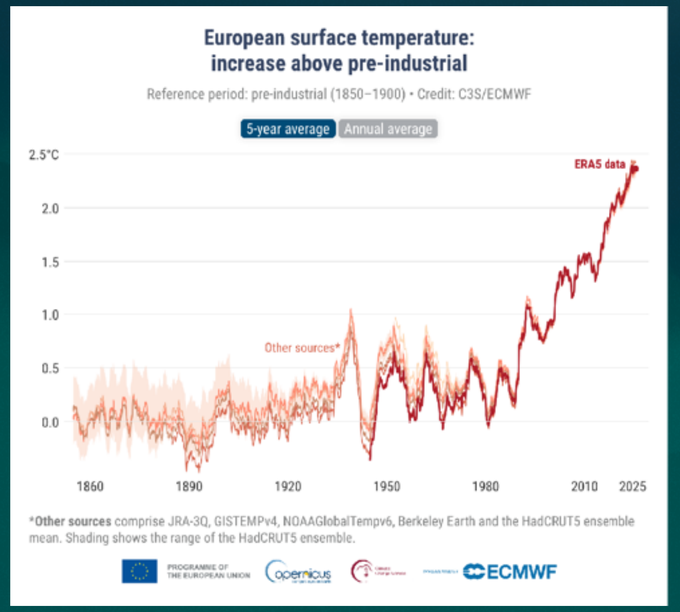
Warming World: A Similar Story from Copernicus/ECMWF

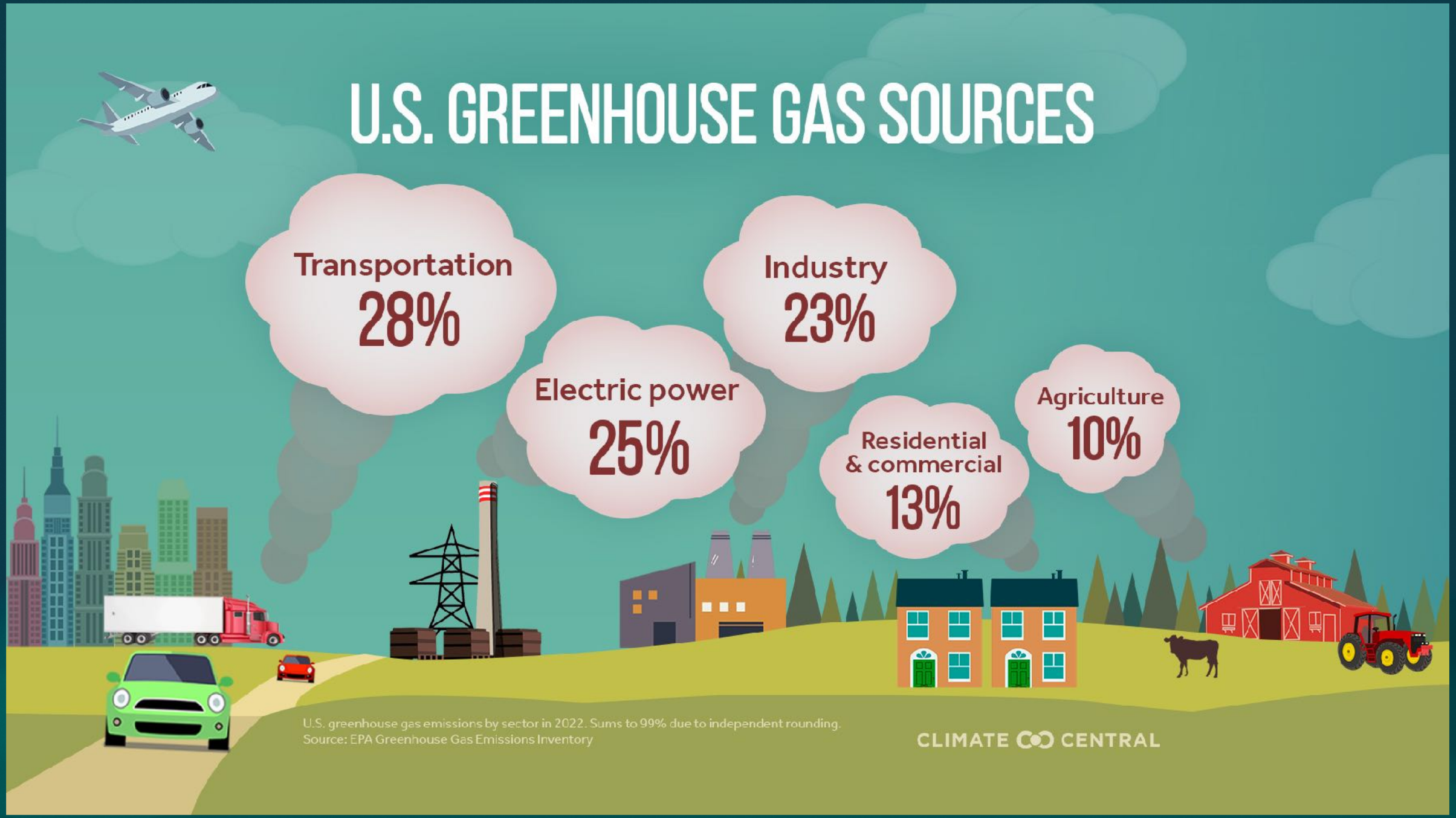








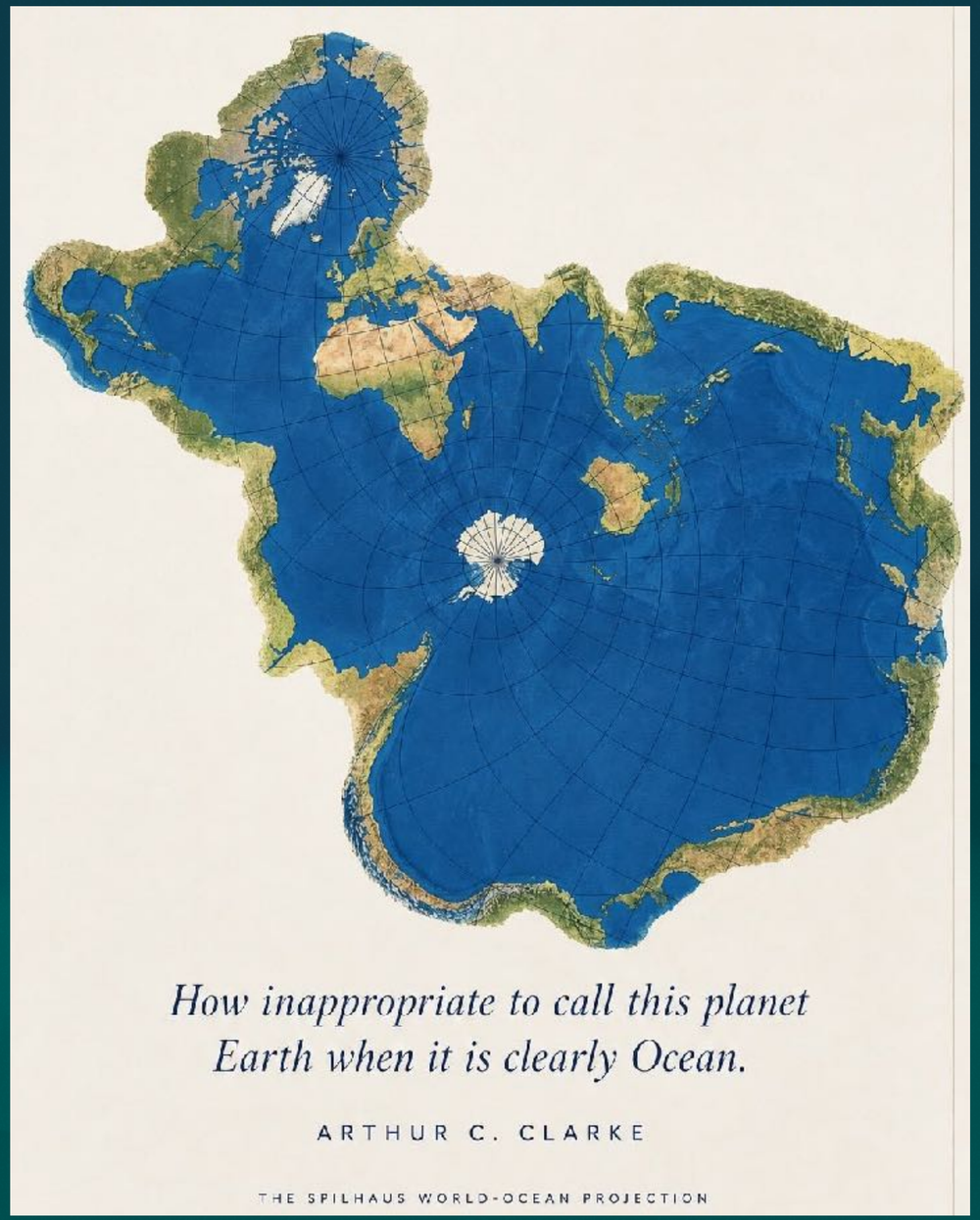


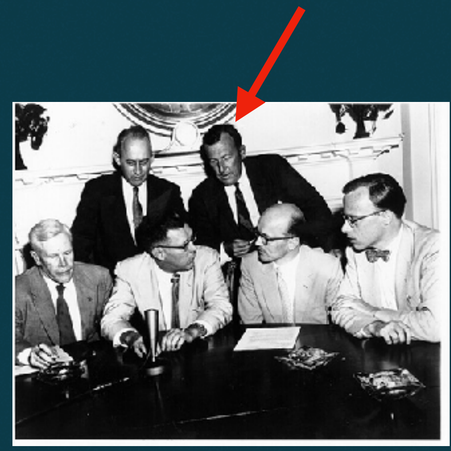


Graphics from Climate Central (Princeton USA)

The Ocean’s Role in Climate

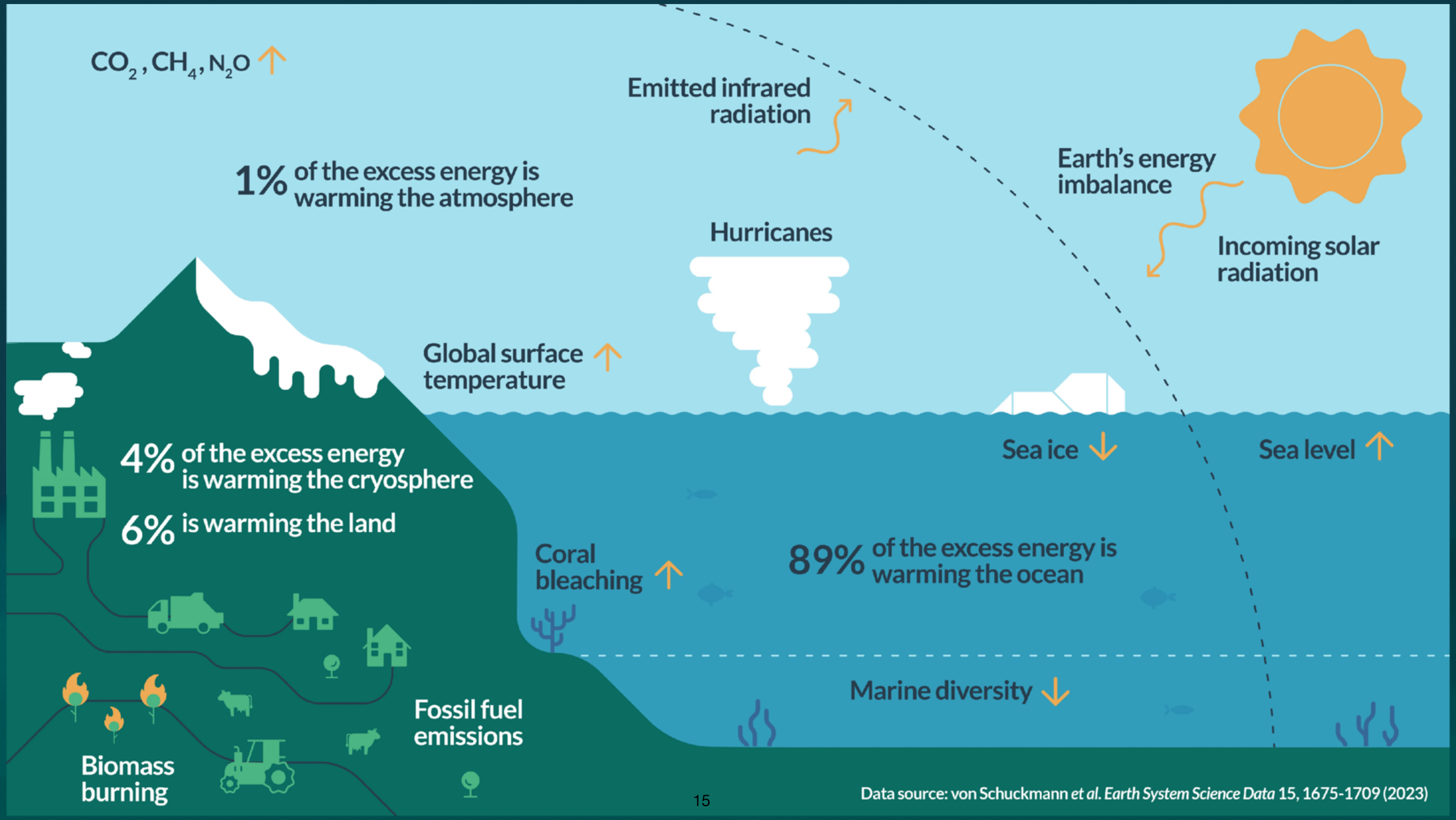
An Ocean World: Spilhaus Map Projection



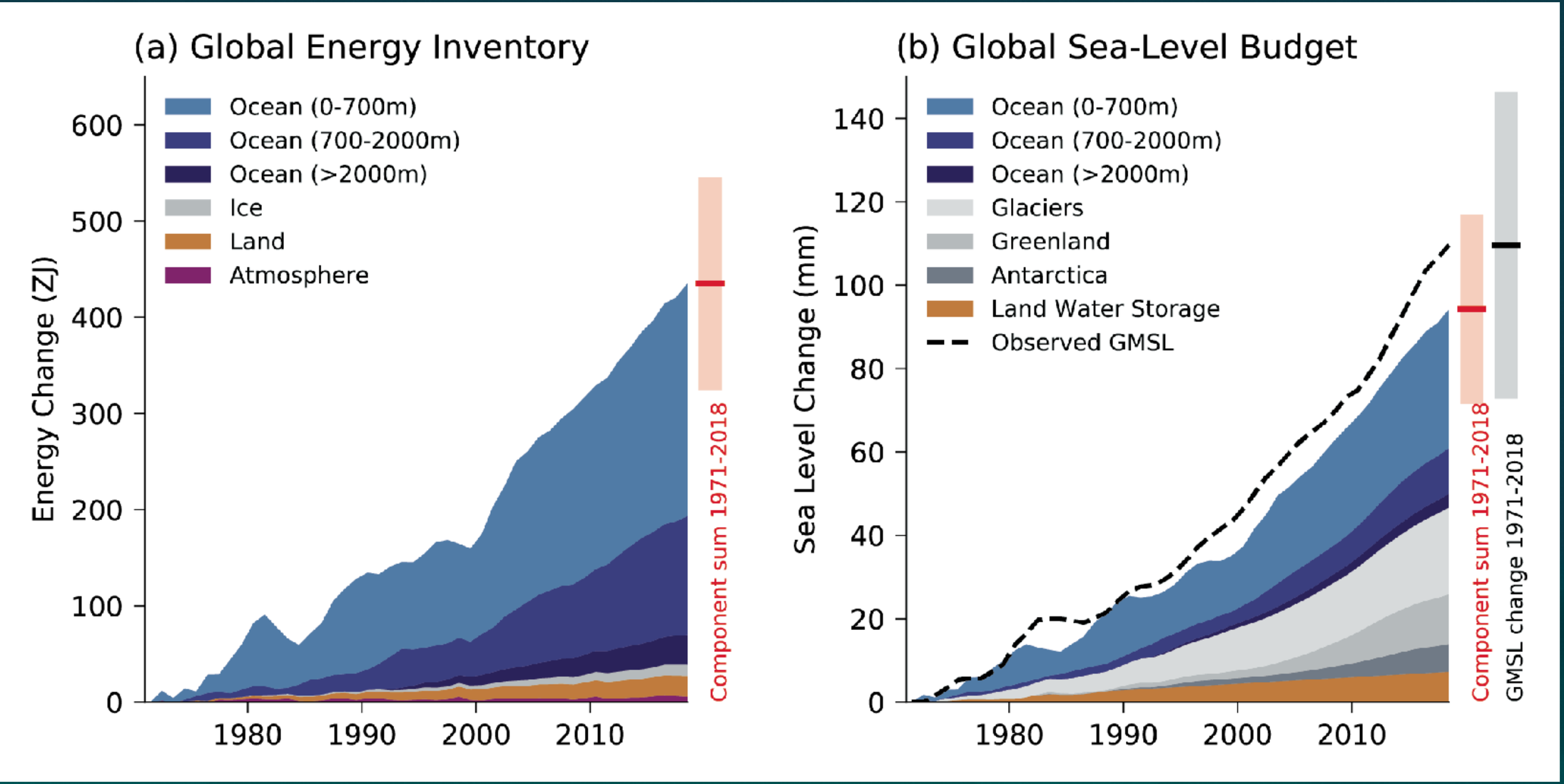


- An ocean-centric cartographic projection depicting the oceans as a single, continuous, interconnected body of water.
- Instead of keeping continents whole and interrupting the oceans, it interrupts and distorts the landmasses to keep the global ocean unbroken.

Global warming is ocean warming

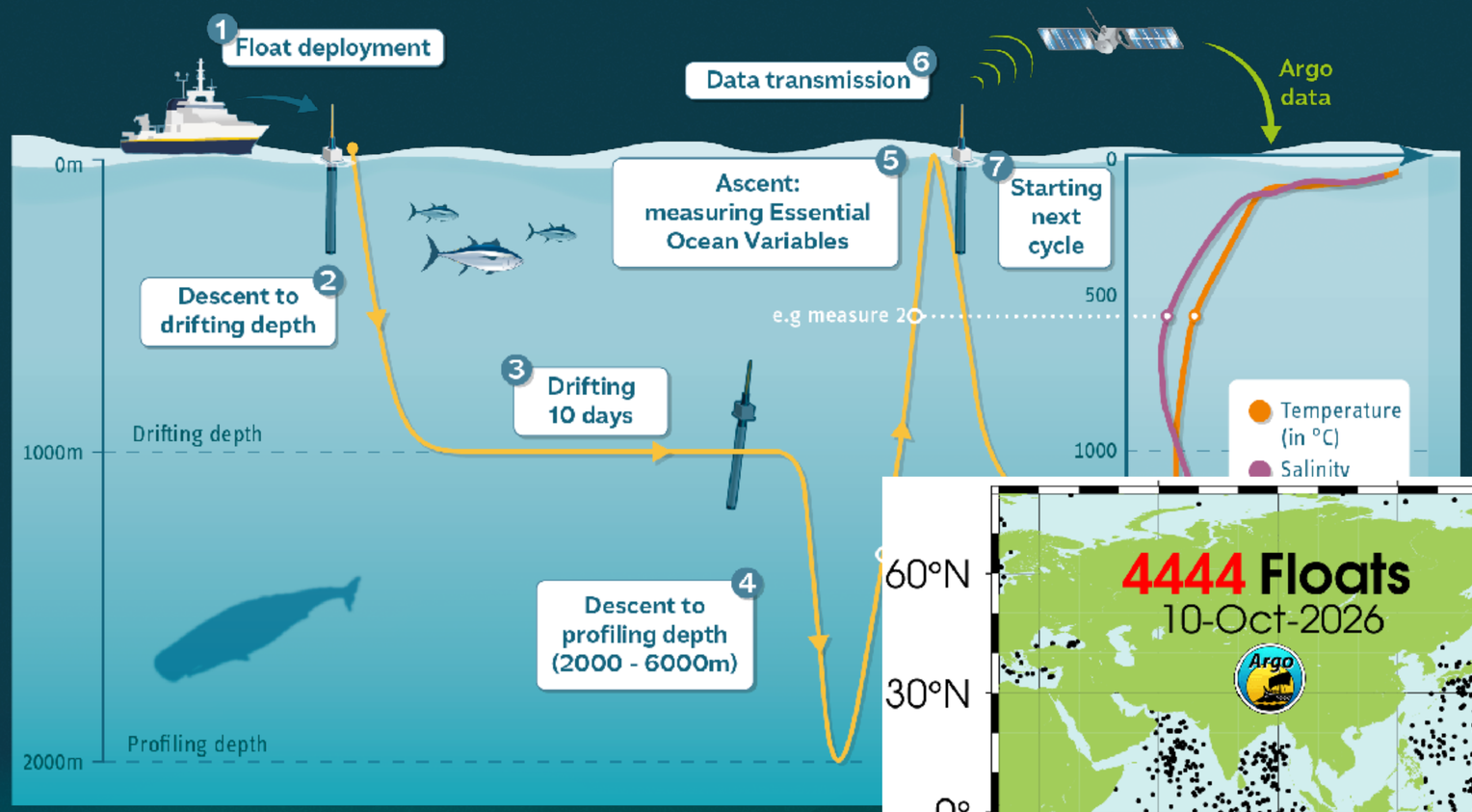


Ocean warming and sea level rise

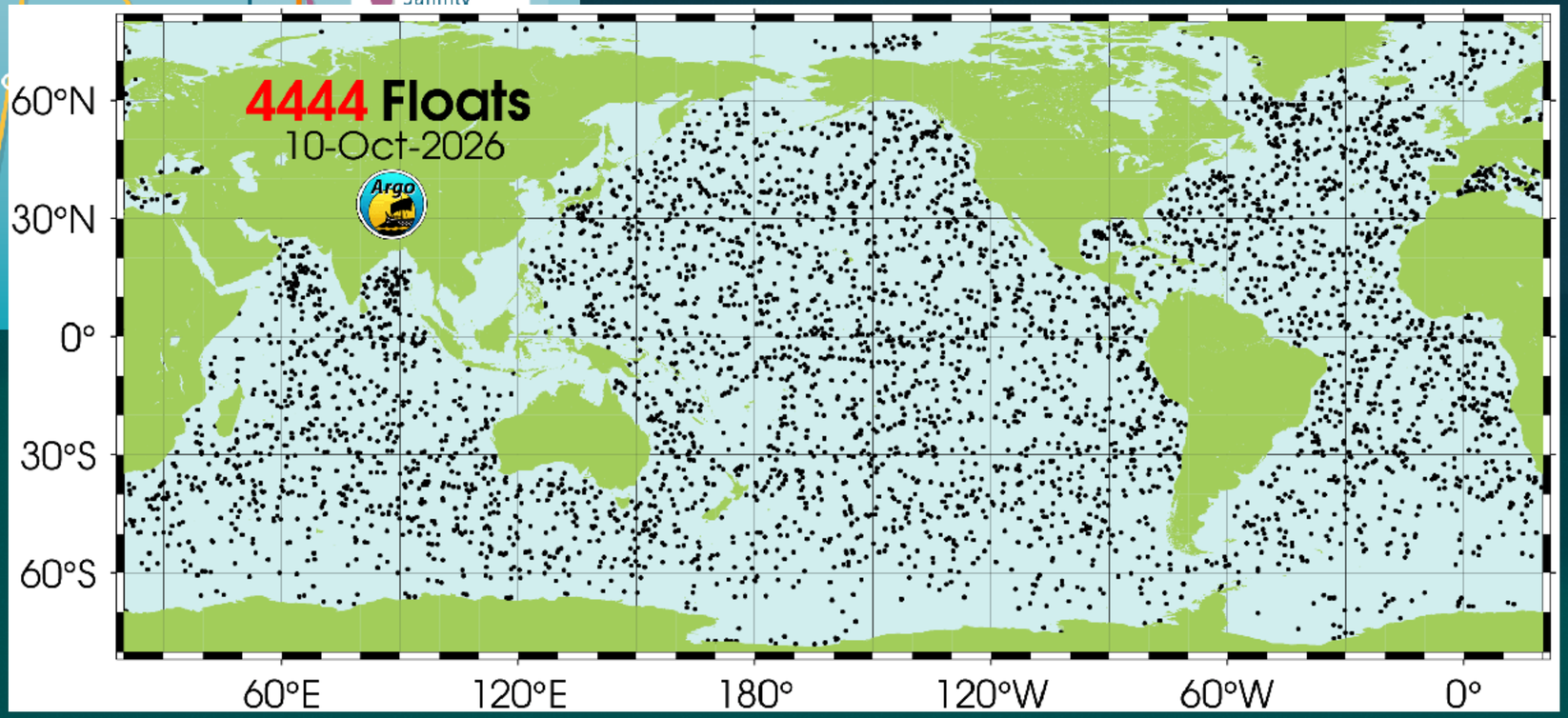


https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-9/

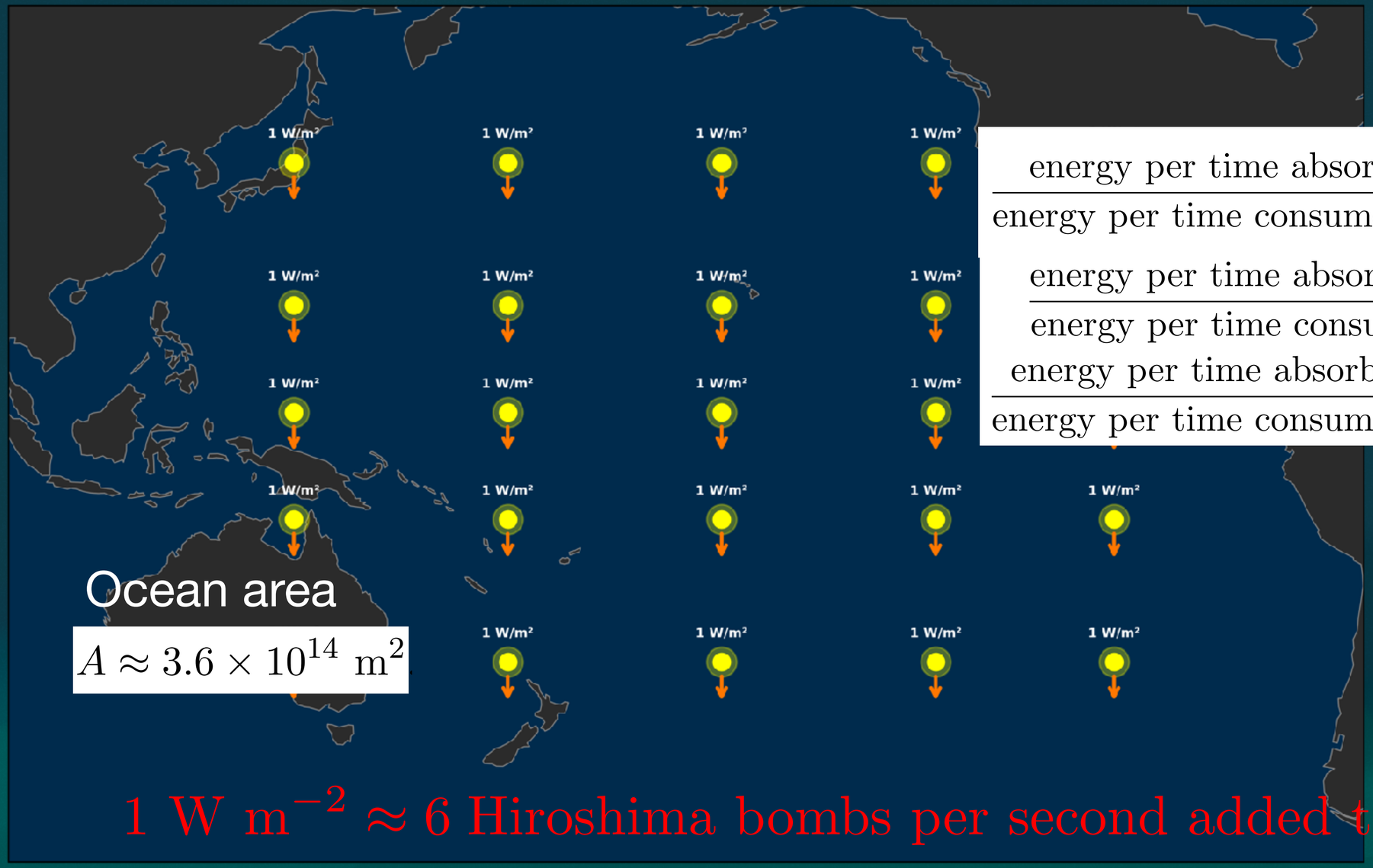
Argo Floats: The Central Tool for Measuring Ocean Warming



Griffies + Legg deploying an Argo float in the Southern Ocean



Magnitudes for the earth energy imbalance: an ocean perspective



$$\frac{\text{energy per time absorbed by ocean}}{\text{energy per time consumed by humanity}} \approx \frac{361\ \text{TW}}{19\ \text{TW}} \approx 19$$

$$\frac{\text{energy per time absorbed by ocean}}{\text{energy per time consumed by USA}} \approx \frac{361\ \text{TW}}{3.34\ \text{TW}} \approx 110$$

$$\frac{\text{energy per time absorbed by ocean}}{\text{energy per time consumed by Europe}} \approx \frac{361\ \text{TW}}{2.54\ \text{TW}} \approx 142$$

$1\ \mathrm{W\ m^{-2}} \approx 6$ Hiroshima bombs per second added to the ocean

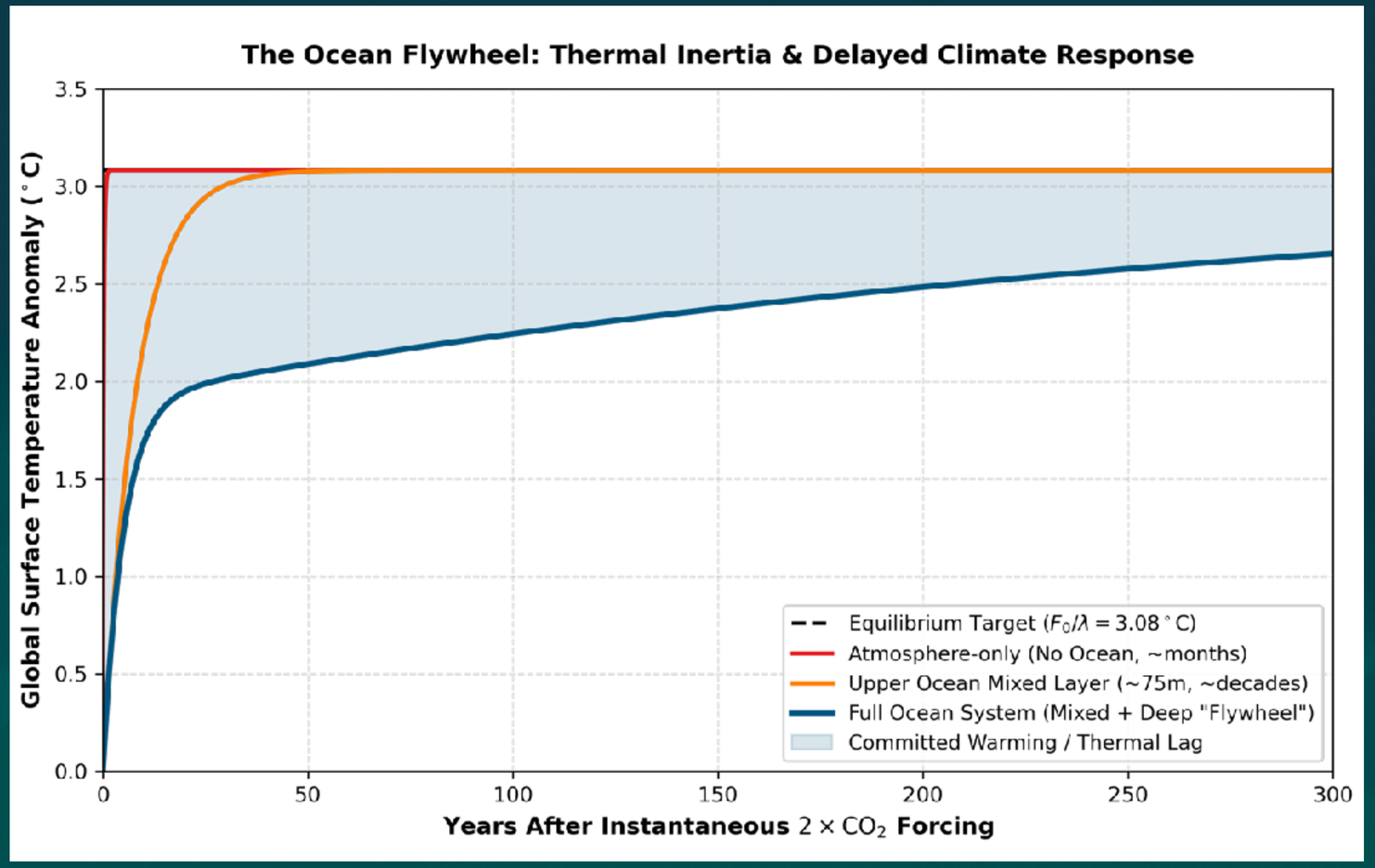
The ocean is the flywheel for Earth's climate

Ocean and atmosphere volumetric heat capacities are very different*

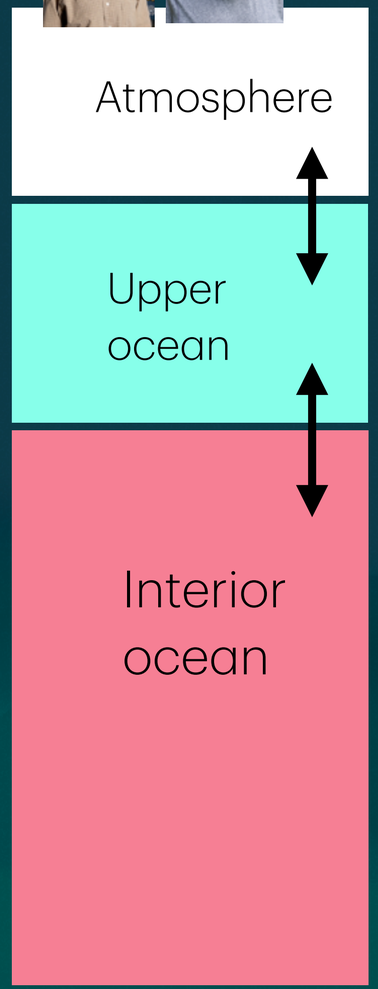
$$\frac{\rho_{\text{ocn}}}{\rho_{\text{atm}}}\frac{C_{\text{pocn}}}{C_{\text{patm}}} \approx \left[\frac{1025\ \text{kg m}^{-3}}{1.225\ \text{kg m}^{-3}}\right]\left[\frac{3990\ \text{J kg}^{-1}\ \text{K}^{-1}}{1004\ \text{J kg}^{-1}\ \text{K}^{-1}}\right] \approx 837 \times 3.97 \approx 3325$$

The ocean has far more ``thermal inertia" than the atmosphere.

2.5 meters of seawater carries the same integrated heat capacity as the full atmospheric column above (about 100 km).



Winton-Held Conceptual Model



*For the experts: ``heat" here mostly refers to differences in enthalpy.

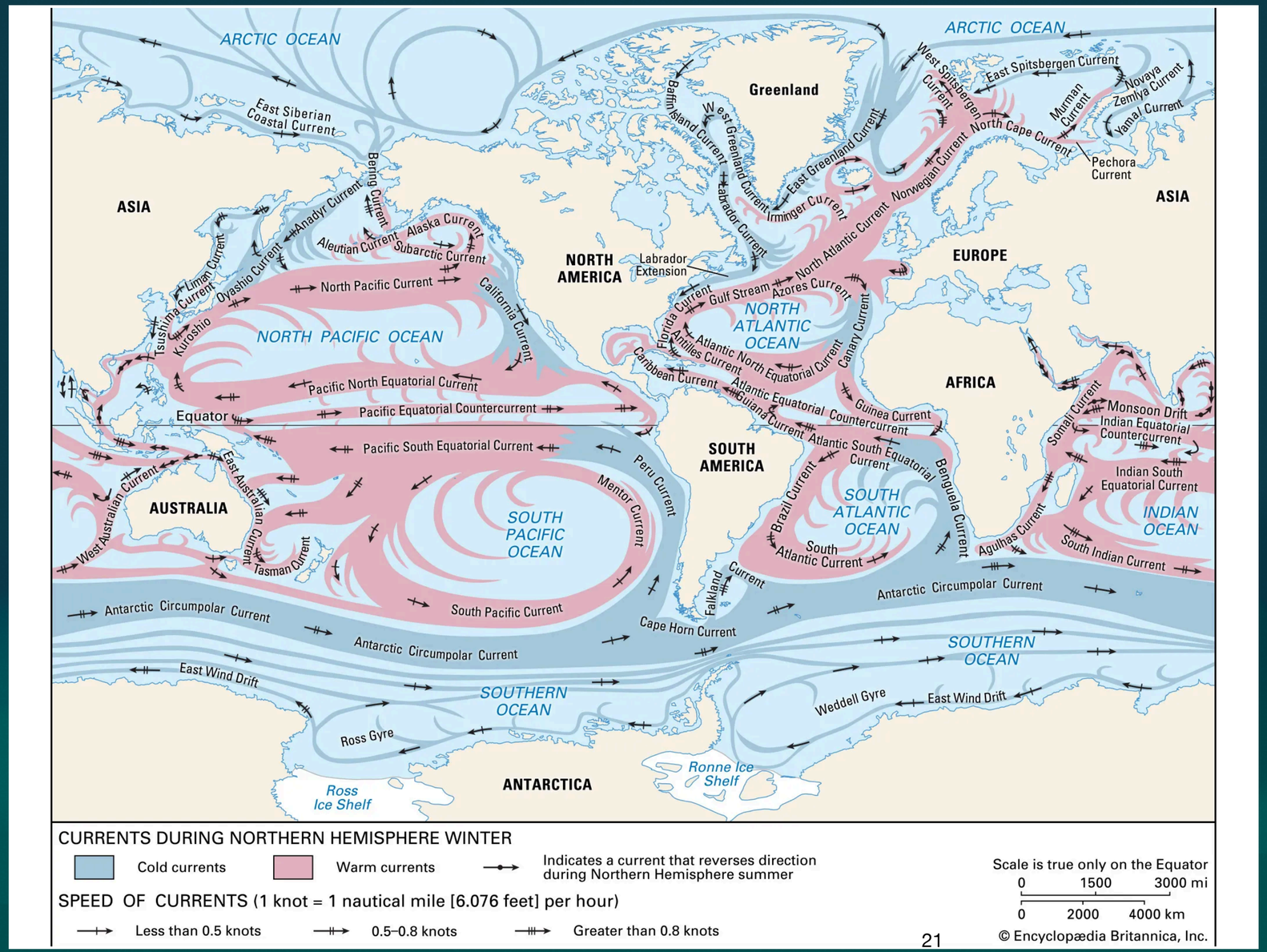
Some Questions about the Ocean and Climate Change

- How does the heat (enthalpy) from the atmosphere get into the ocean?
- What happens to that heat once it gets into the ocean?
- Will the heat stay in the ocean forever (``out of site, out of mind")?
- Will heat resurface at another time and place (``remember me?").

- Ocean physics is central to answering these questions.
- Motion and its cause via forces (Newton's Laws) describe ocean currents.
- Thermal processes are described by the principles of thermodynamics.
- Physicists aim to answer ``Why Questions" arising from observations.
- Physics is the foundation for computer models used to study + predict climate.

The large-scale ocean circulation

The ocean is not a swamp! It is a zoo of physical processes that circulates, stirs, and mixes!



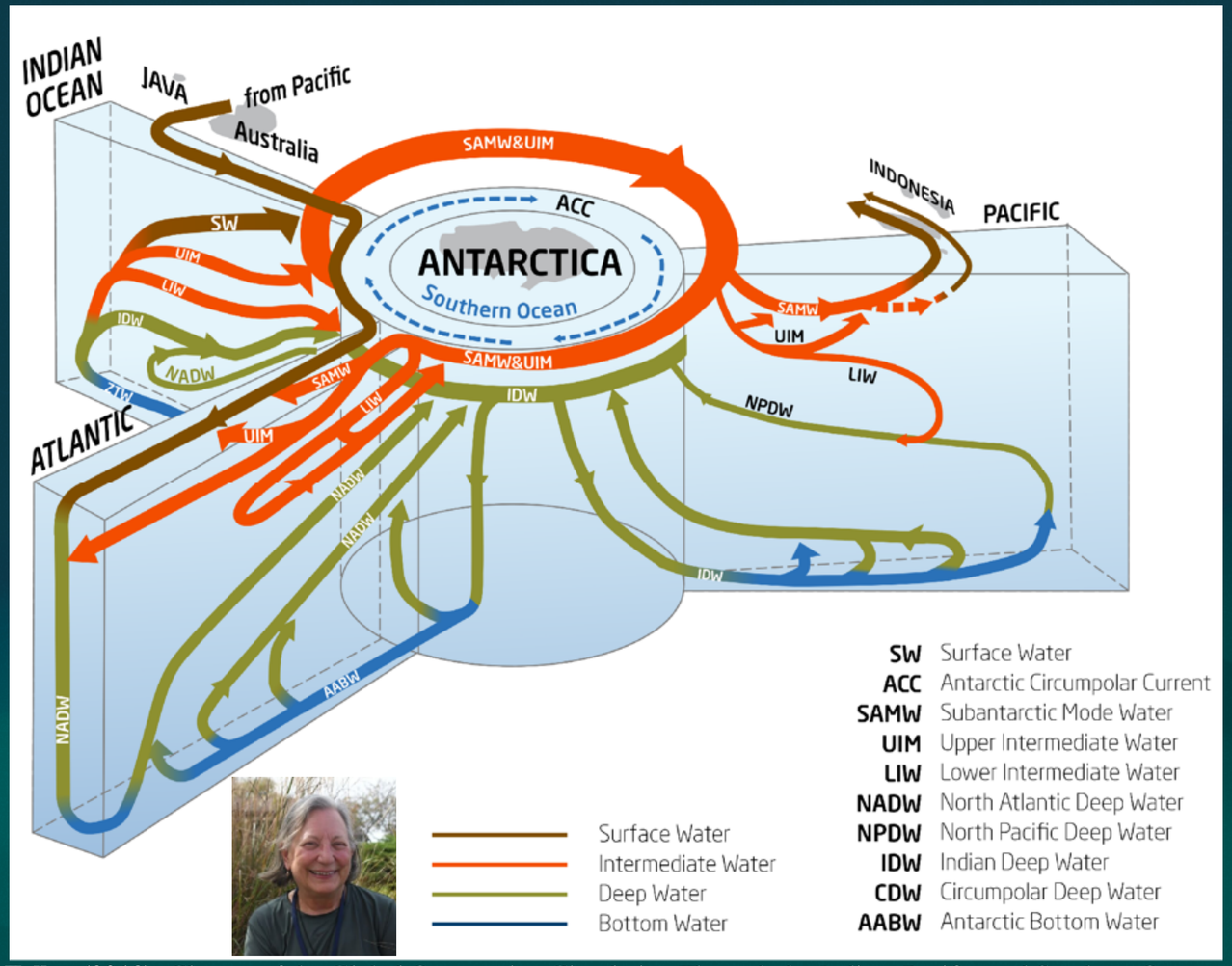
Why does the ocean have such circulation patterns?

What causes these currents to flow?

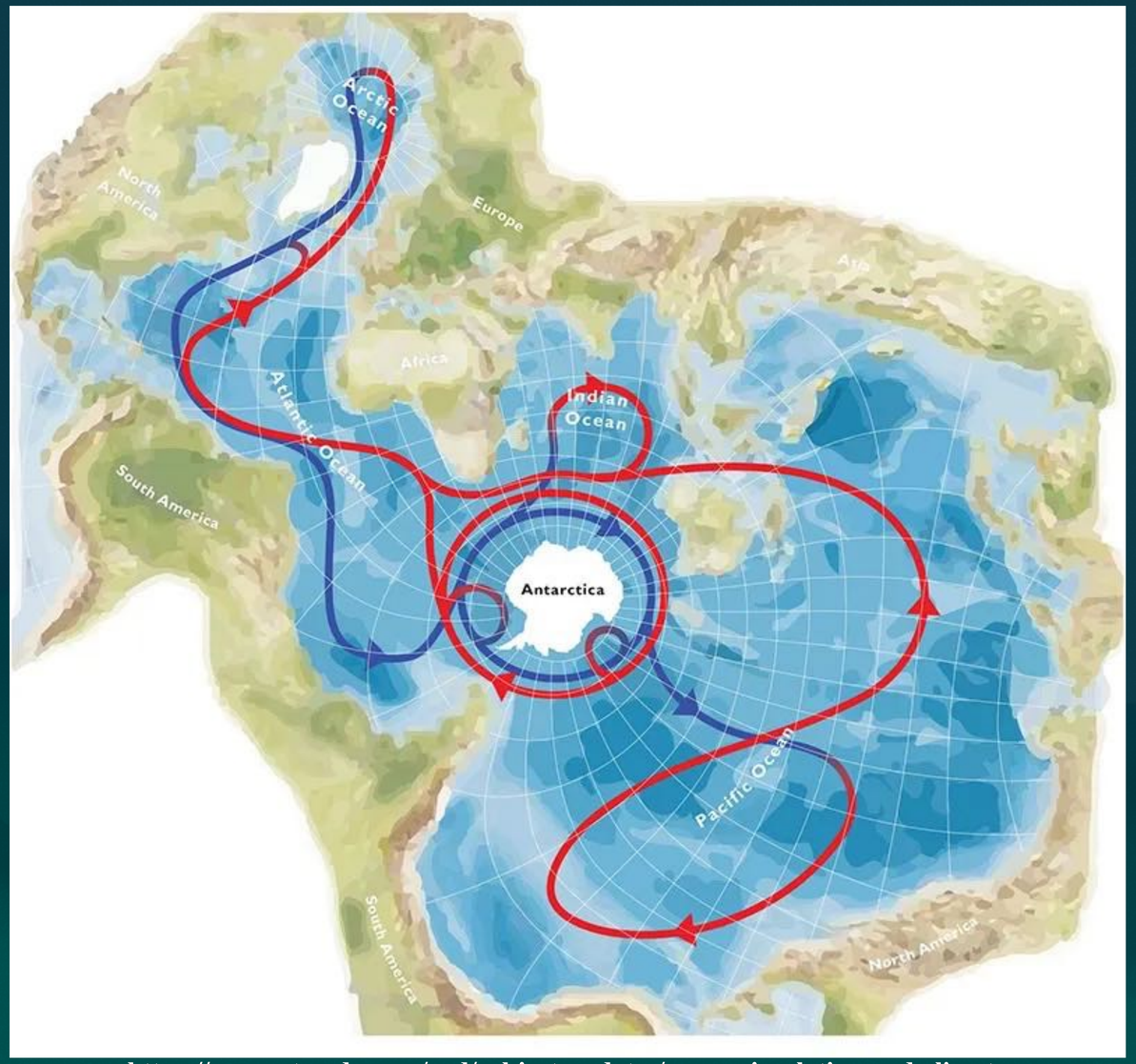
These questions are central to the study of ocean physics.

The Ocean Overturning Circulation

Wind and buoyancy forcing act to create motion in the vertical and horizontal directions that create giant overturning cells (ocean “plumbing”) .

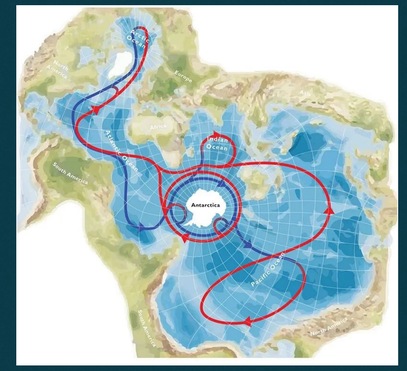


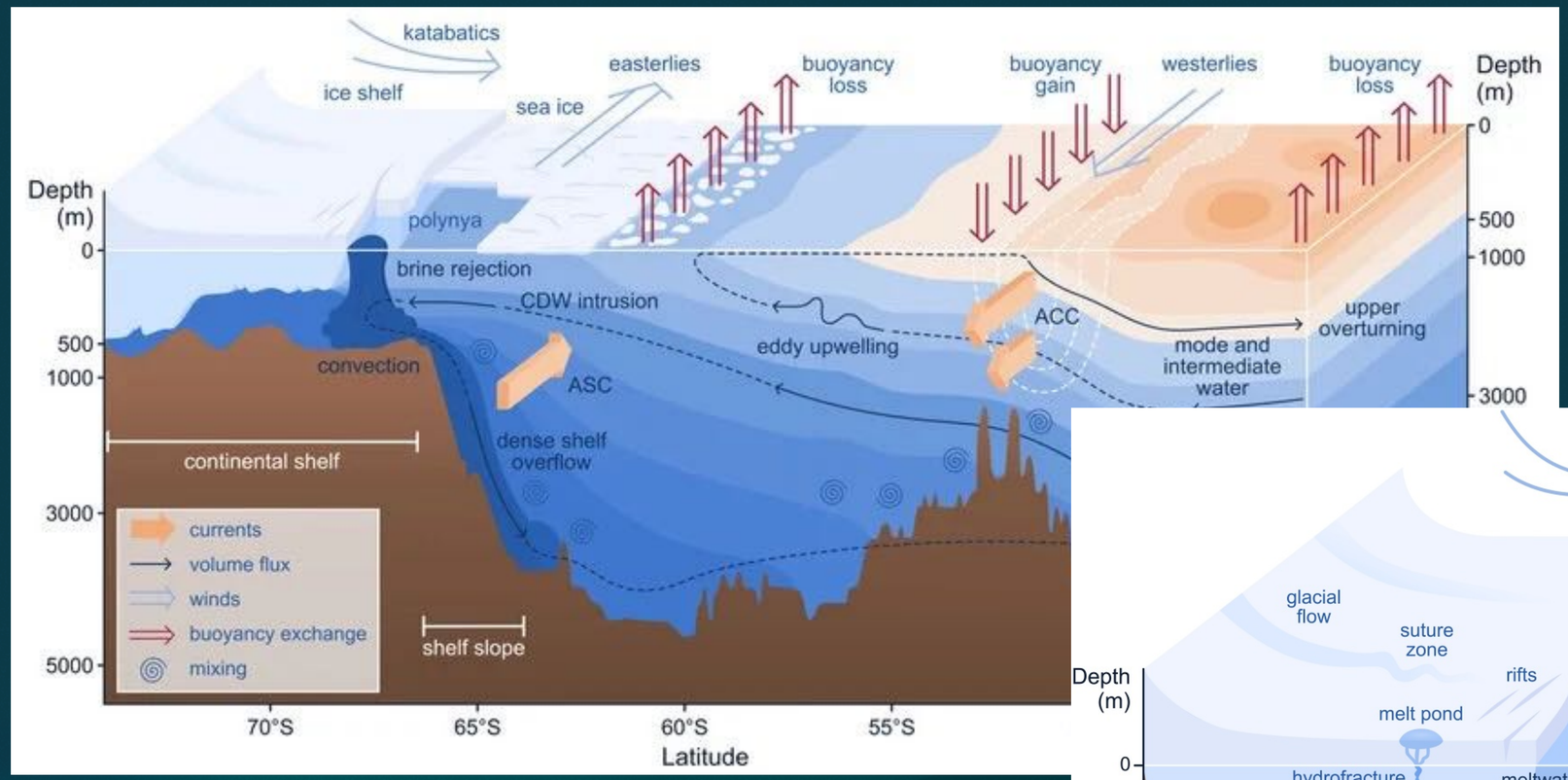
Talley (2013): Closure of the Global Overturning Circulation Through the Indian, Pacific, and Southern Oceans.

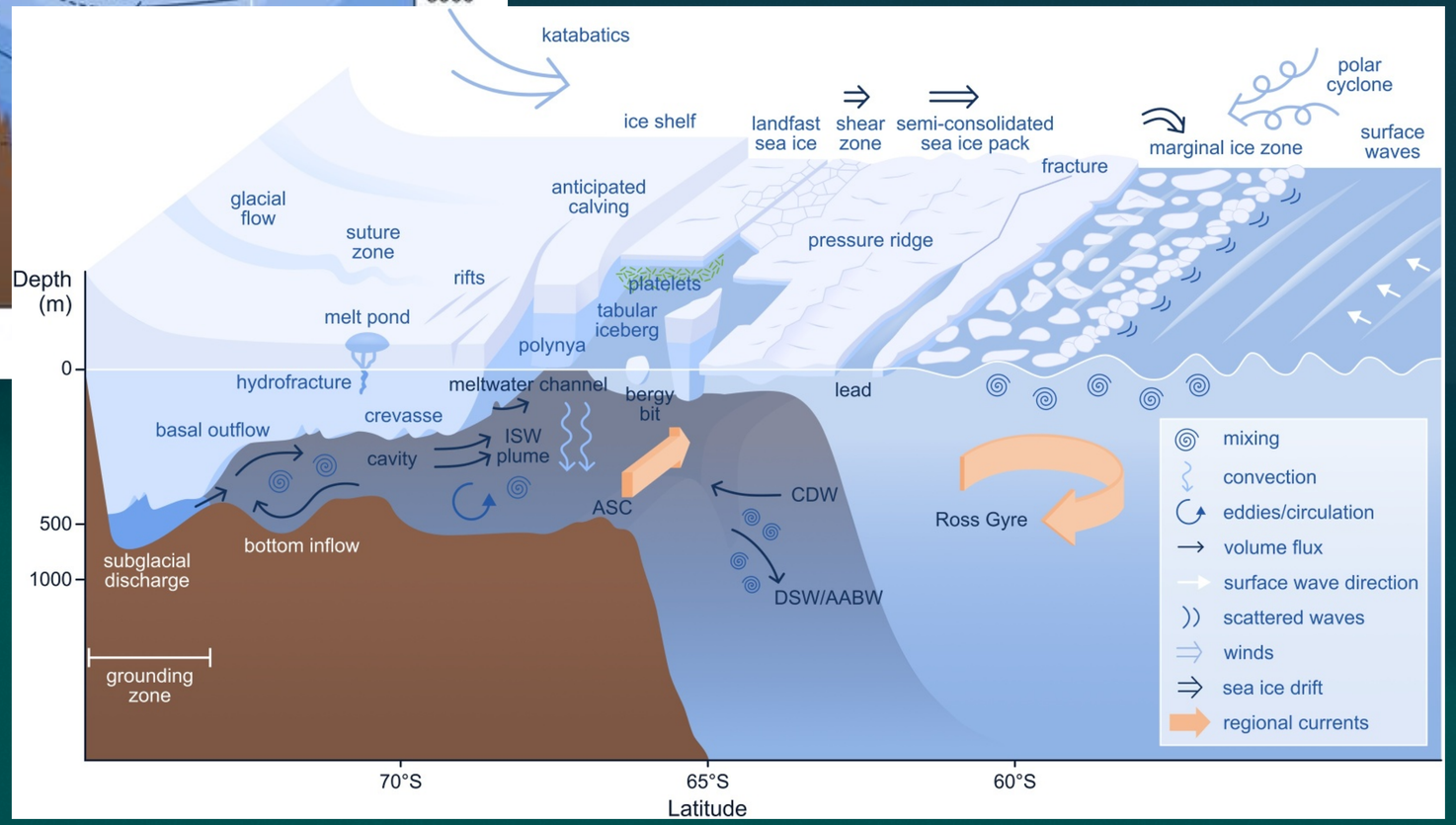


https://encounteredu.com/cpd/subject-updates/ocean-circulation-and-climate

A Focus on the Southern Ocean







Bennetts et al 2024, Reviews of Geophysics

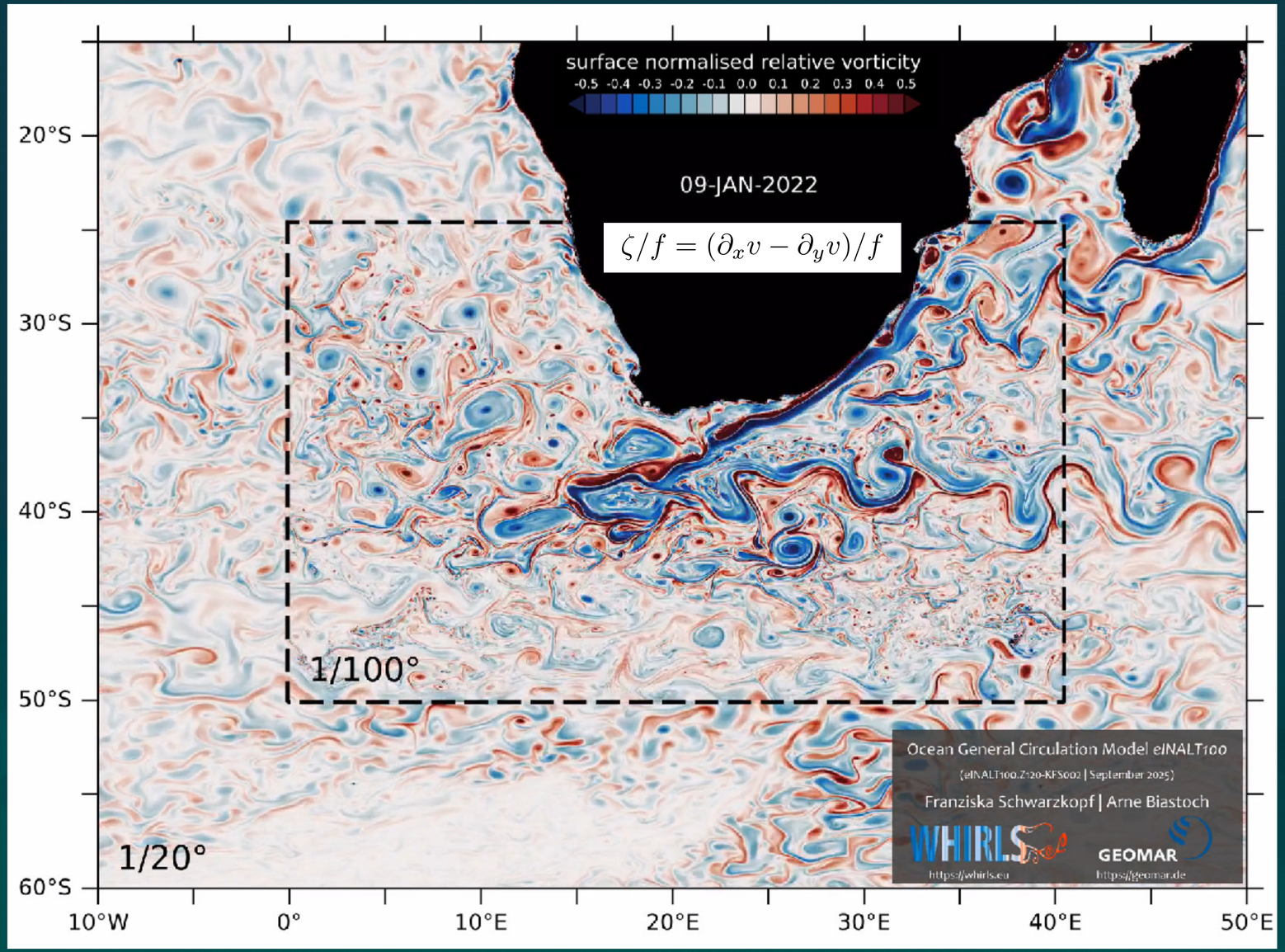
Multiple scales of ocean flows

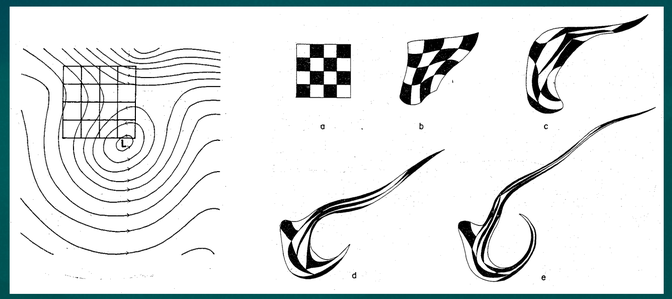
Sea surface temperature from a computer simulation

(click for animation) Ocean surface temperatures from a NOAA/GFDL coupled climate model CM2.6.

Normalized relative vorticity from eINALT100 simulation

Click for animation

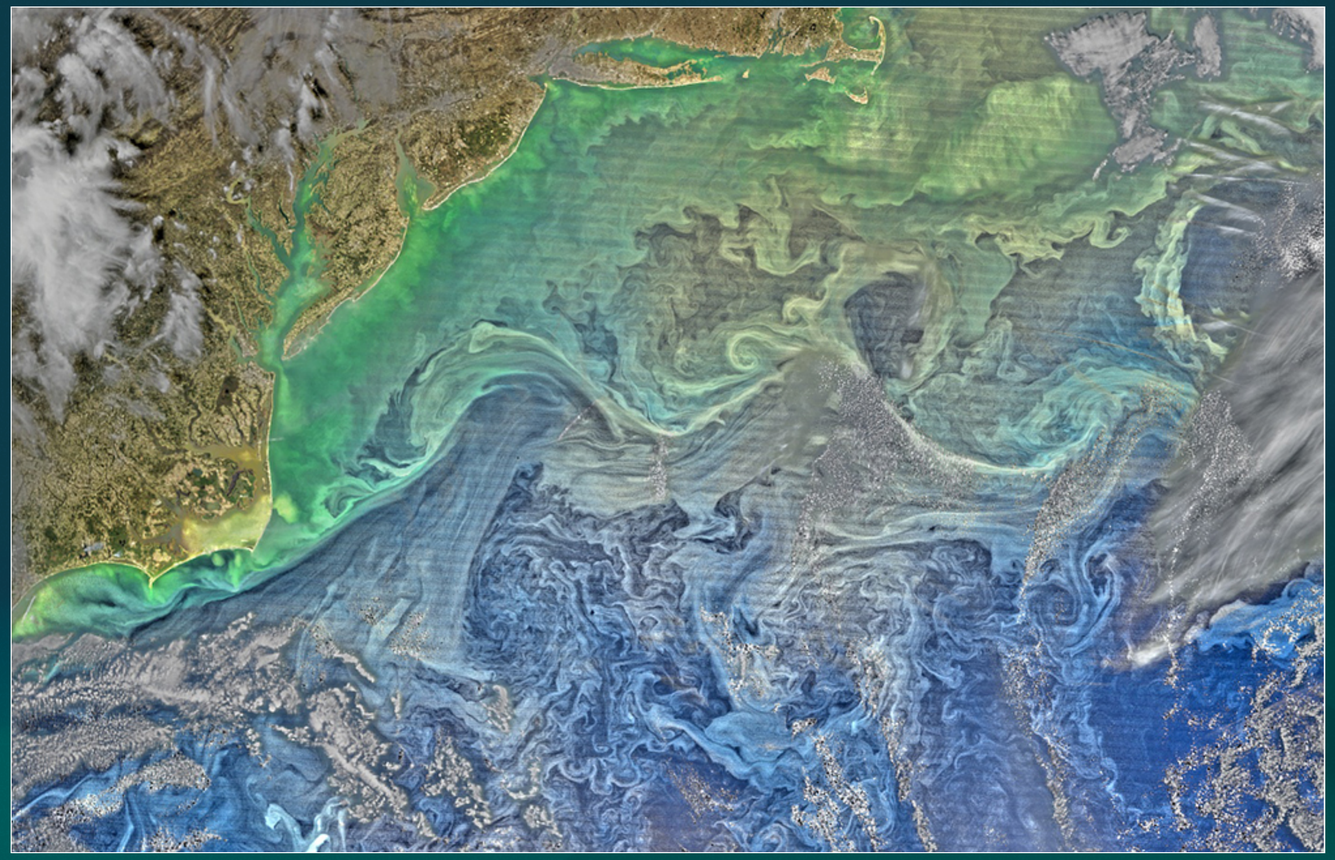




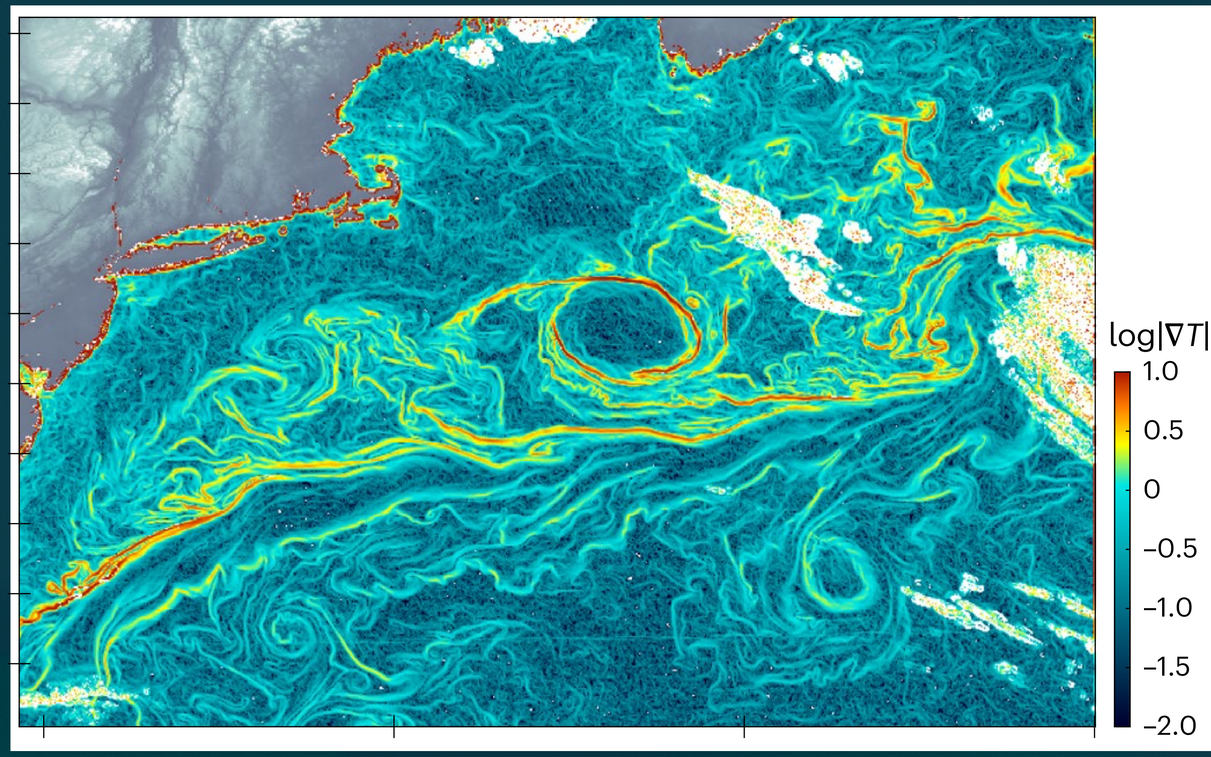
Welander 1955

Thanks to Rene Schubert, GEOMAR, Kiel, Germany

Satellite images of multiple ocean scales



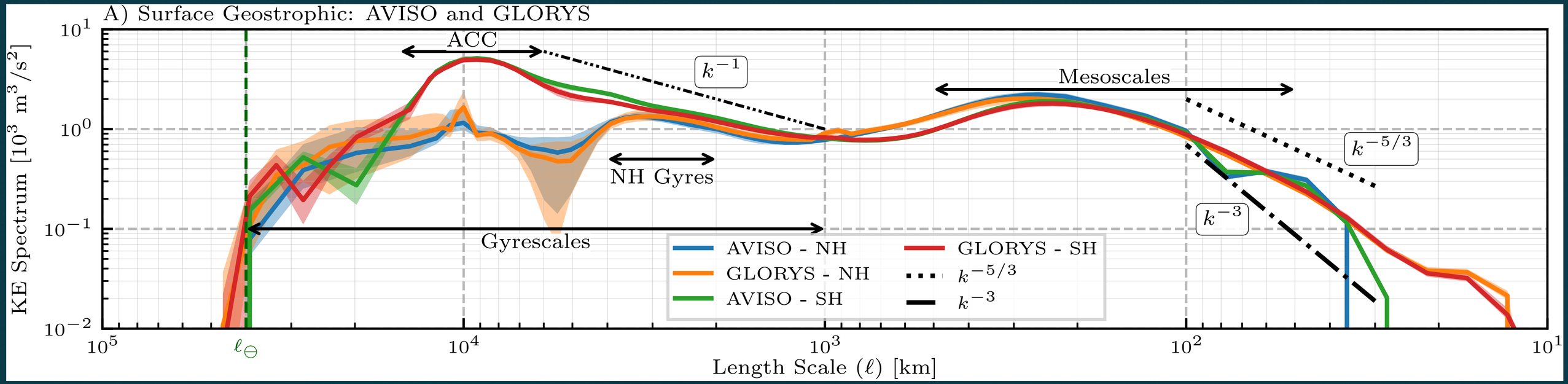
https://www.nasa.gov/image-article/turbulent-north-atlantic/ : turbulence made visible with phytoplankton

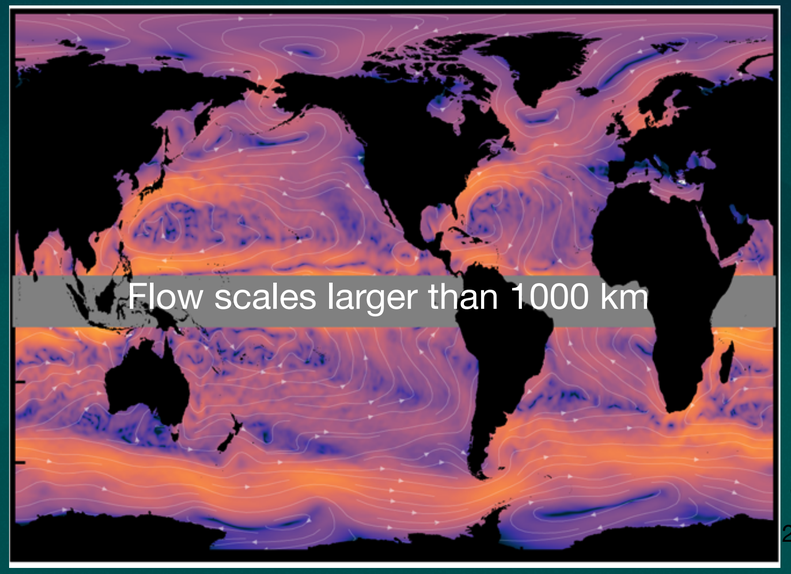


Lenain et al 2026: |ln \nabla T| estimated via ML

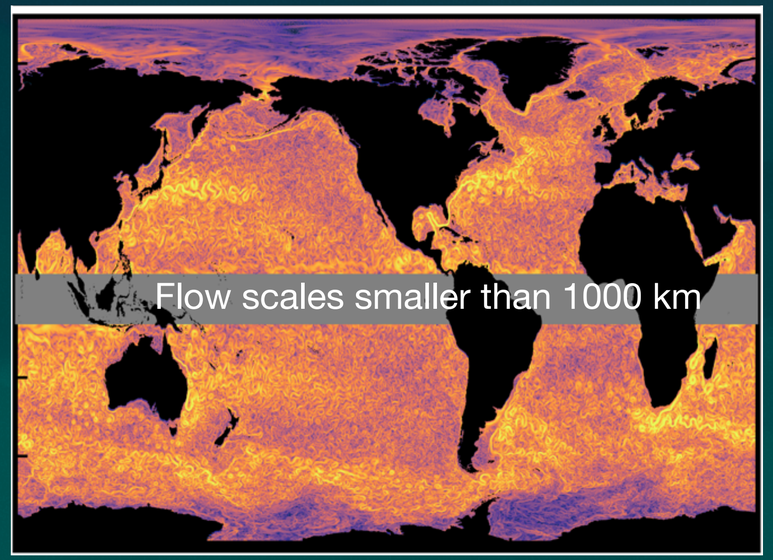
Spectral decomposition of the ocean kinetic energy

Aluie et al 2027

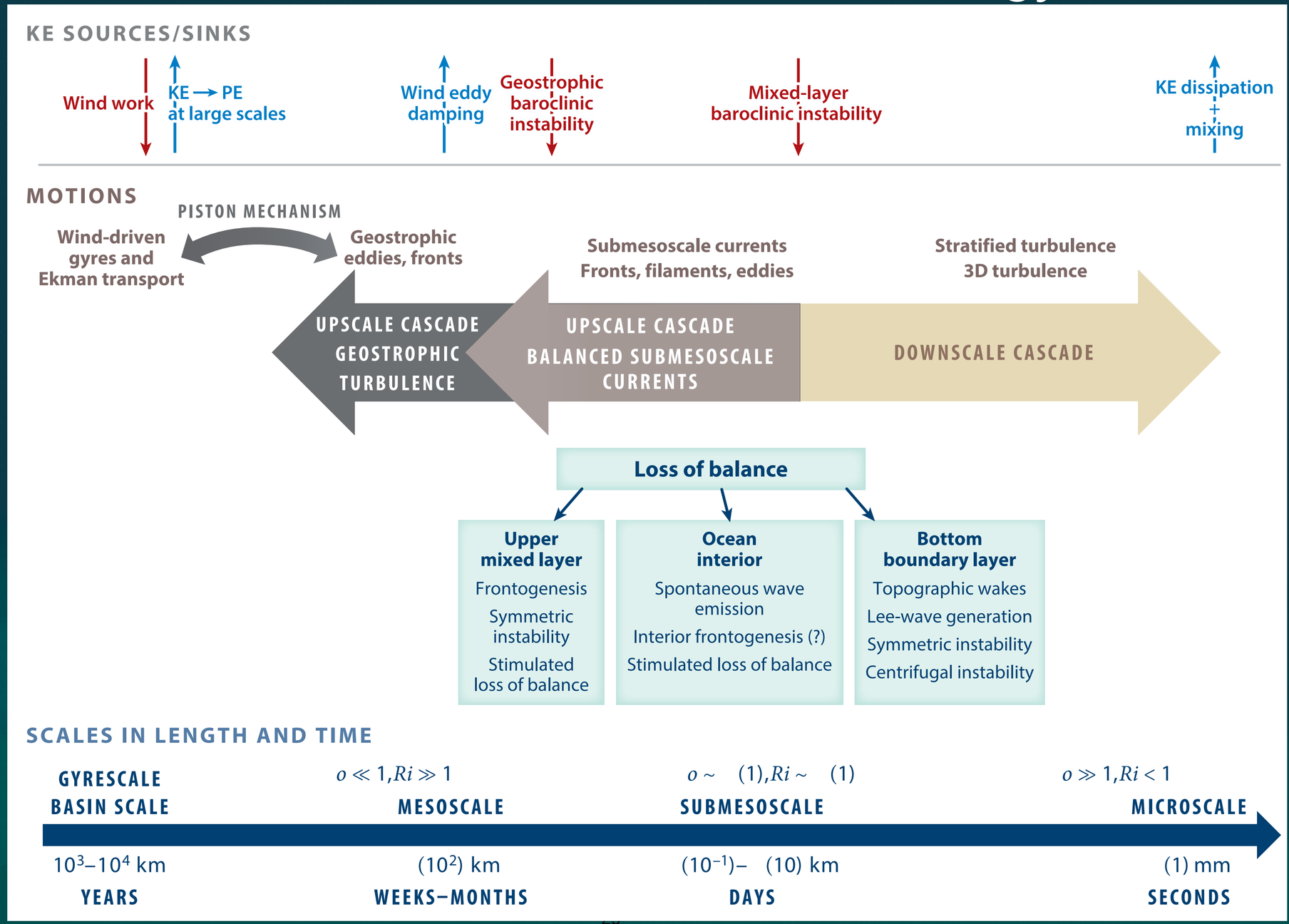


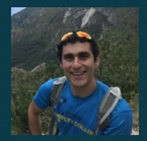


Storer et al 2022



Mechanisms for the Transfer of Kinetic Energy Across Scales

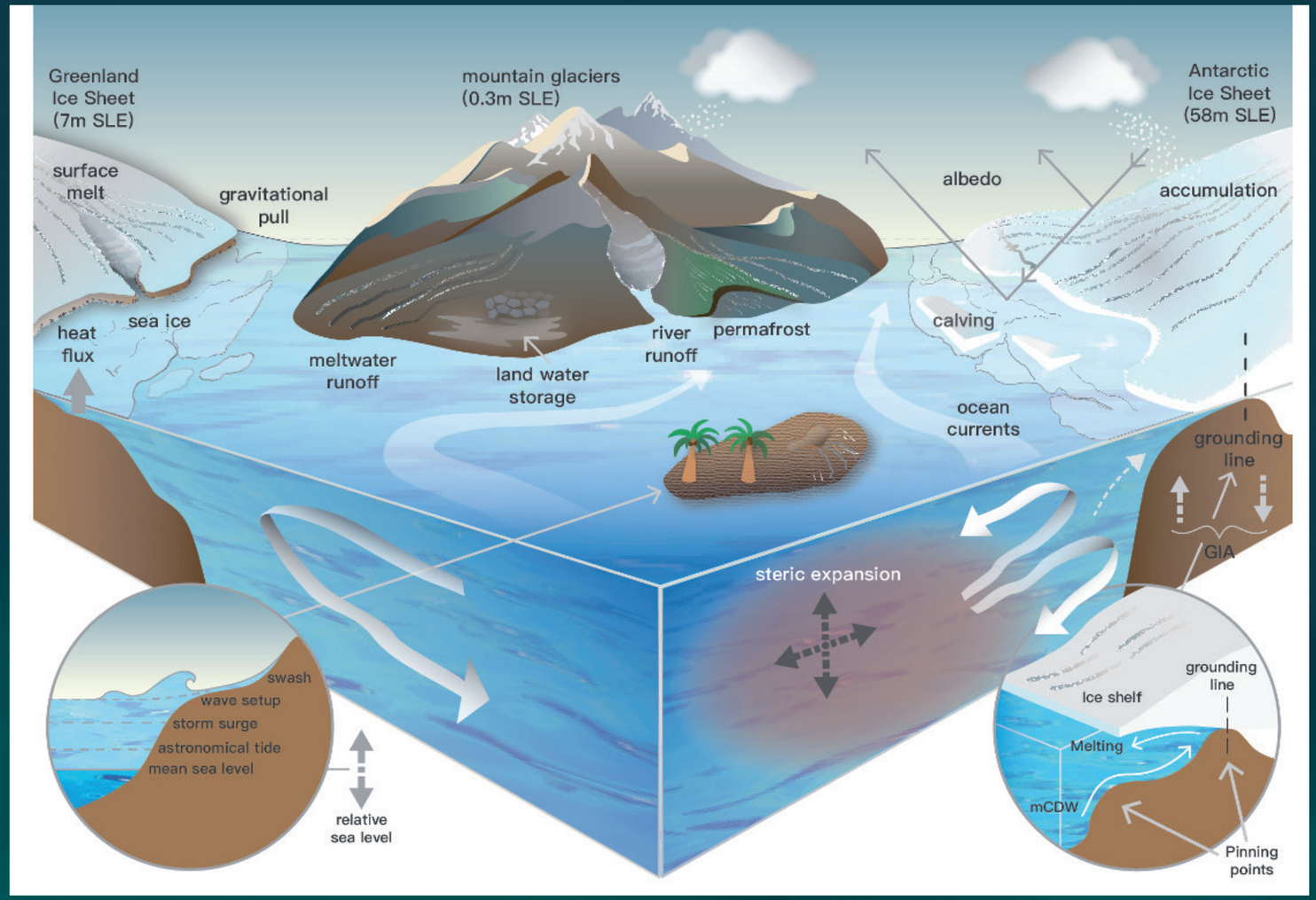




Aluie et al 2027

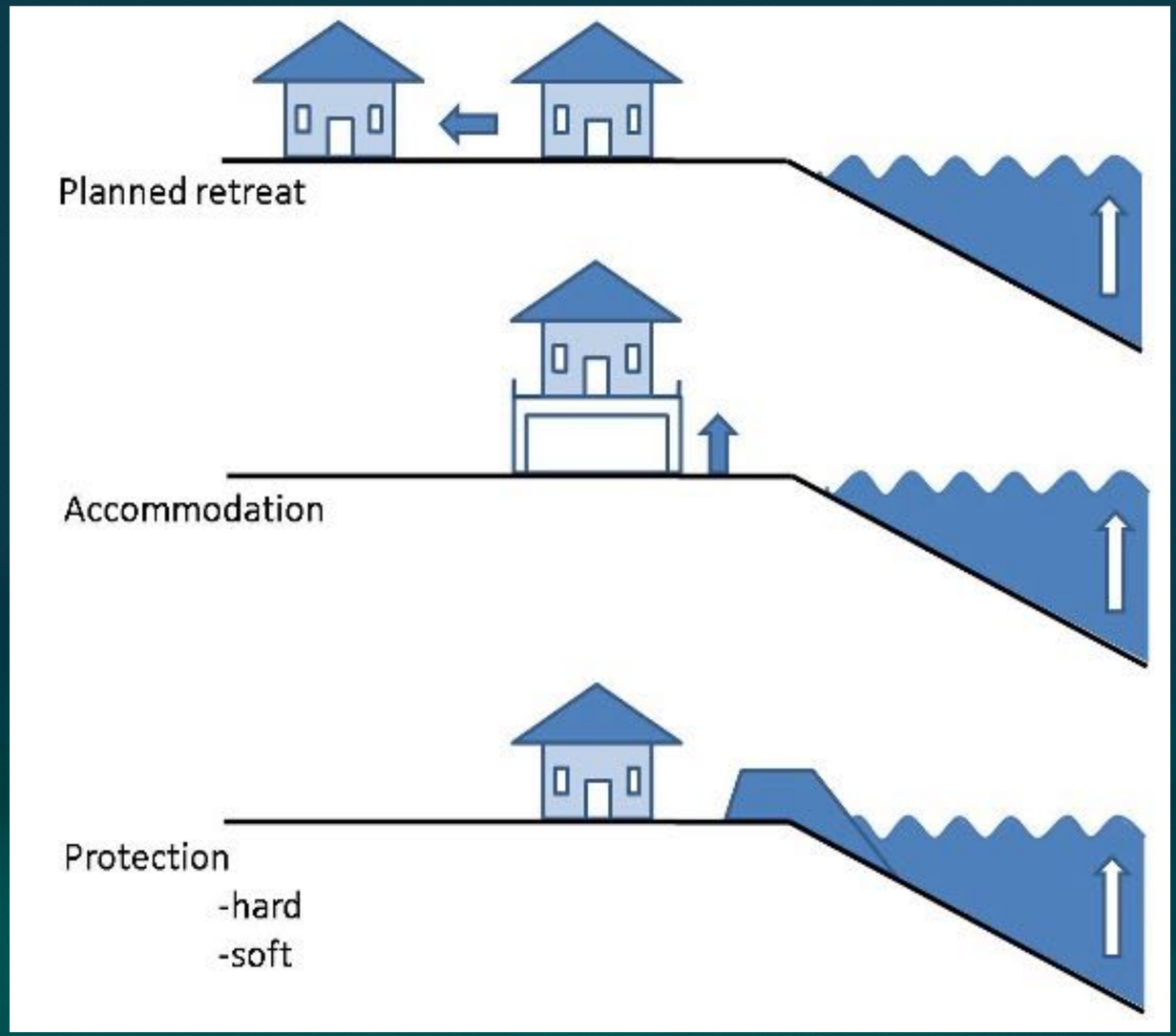
Sea Level Rise: Global and Regional

Contributors to Sea Level Rise



IPCC_AR6_WGI_Chapter09

Sea Level Rise: The Need to Adapt



Nicholls et al. 2010

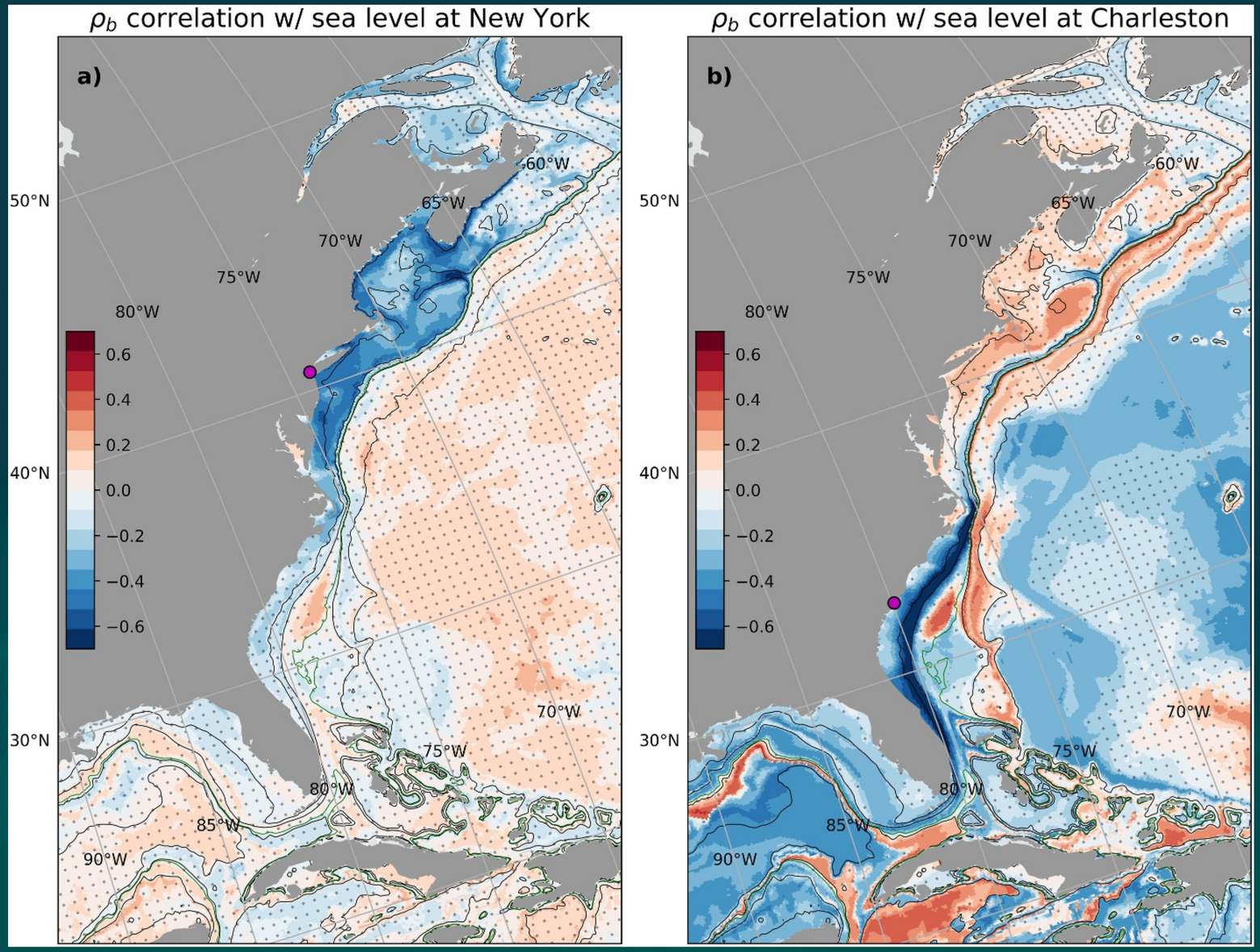
Sea Level Rise: A Moving Coast

Many Factors to Consider:

- Tides
- Storm Surge
- Waves
- Seasonal to Interannual Variations
- Sea Level Rise

https://ncseagrant.ncsu.edu/coastwatch/seven-more-feet/

Connecting coastal + regional sea level change on US East Coast

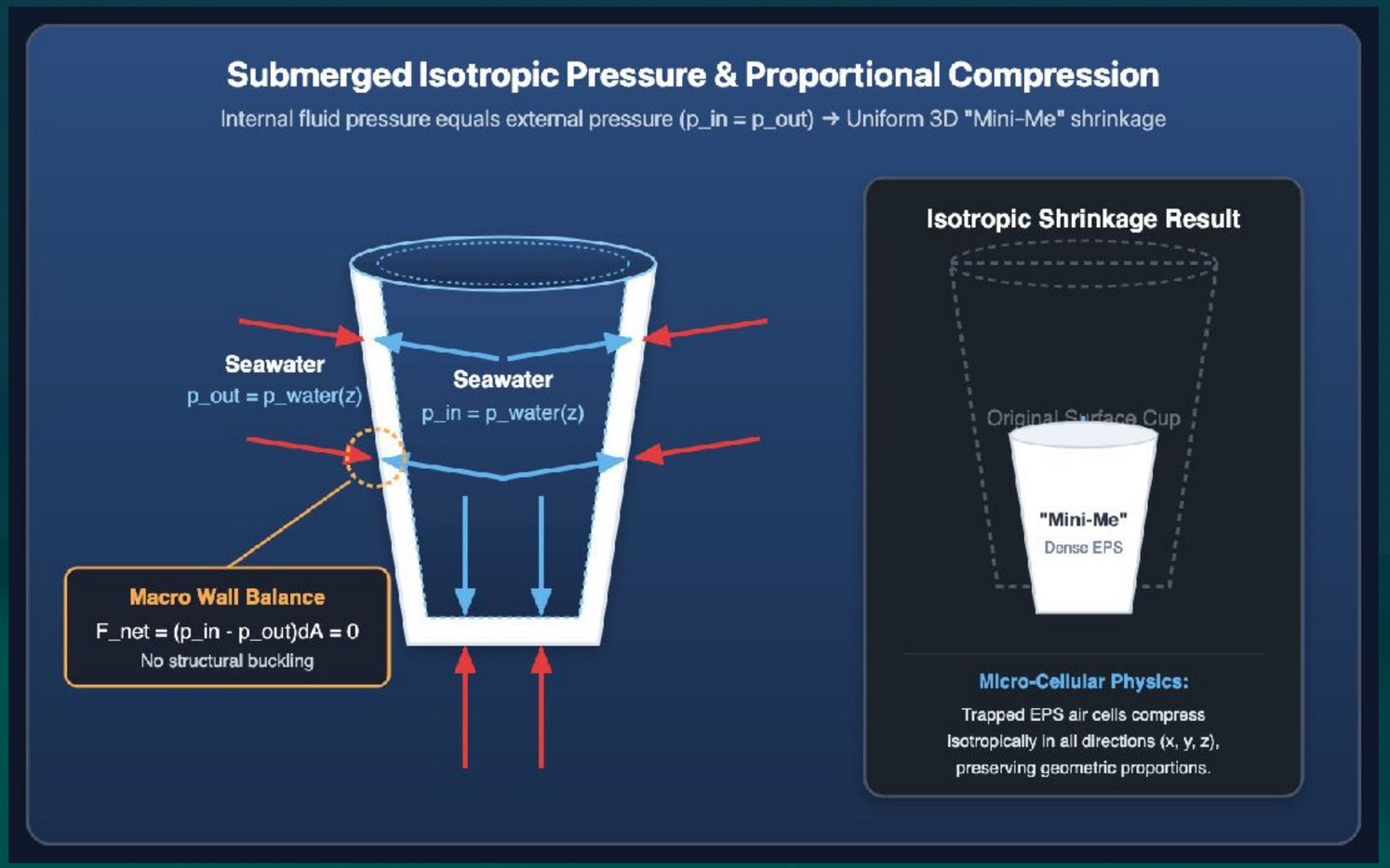


Steinberg, Griffies, Piecuch, 2026: Steric setup effects on US East coast sea level

Some Words about Physics in the Ocean

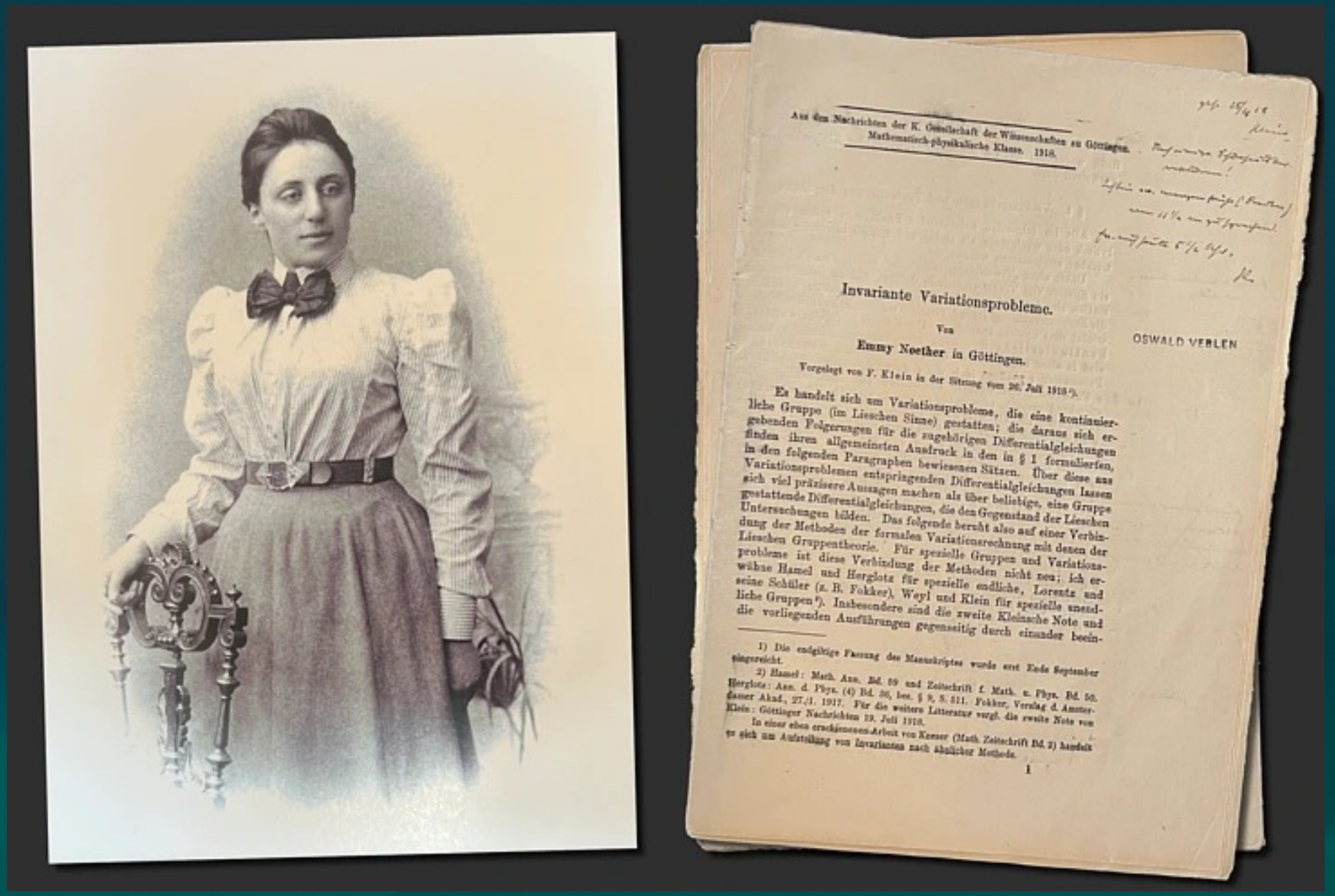
Pressure and the styrofoam cup experiment

Pressure at 5000m is about 500x sea level pressure



Graphic generated by Gemini-AI

Emmy Noether: Connecting Symmetries et Conservation Laws



Aus den Nachrichten der K. Gesellschaft der Wissenschaften zu Göttingen.
Mathematisch-physikalische Klasse. 1918.

Invariante Variationsprobleme.

Von

OSWALD VEBLEN

Emmy Noether in Göttingen.

Vorgelegt von F. Klein in der Sitzung vom 26. Juli 1918[1].

Es handelt sich um Variationsprobleme, die eine kontinuierliche Gruppe (im Lieschen Sinne) gestatten; die daraus sich ergebenden Folgerungen für die zugehörigen Differentialgleichungen finden ihren allgemeinsten Ausdruck in den in § 1 formulierten, in den folgenden Paragraphen bewiesenen Sätzen. Über diese aus Variationsproblemen entspringenden Differentialgleichungen lassen sich viel präzisere Aussagen machen als über beliebige, eine Gruppe gestattende Differentialgleichungen, die den Gegenstand der Lieschen Untersuchungen bilden. Das folgende beruht also auf einer Verbindung der Methoden der formalen Variationsrechnung mit denen der Lieschen Gruppentheorie. Für spezielle Gruppen und Variationsprobleme ist diese Verbindung der Methoden nicht neu; ich erwähne Hamel und Herglotz für spezielle endliche, Lorentz und seine Schüler (z. B. Fokker), Weyl und Klein für spezielle unendliche Gruppen[2]. Insbesondere sind die zweite Kleinsche Note und die vorliegenden Ausführungen gegenseitig durch einander beein-

1) Die endgiltige Fassung des Manuskriptes wurde erst Ende September eingereicht.

2) Hamel: Math. Ann. Bd. 59 und Zeitschrift f. Math. u. Phys. Bd. 50. Herglotz: Ann. d. Phys. (4) Bd. 36, bes. § 9, S. 511. Fokker, Verslag d. Amsterdamer Akad., 27./1. 1917. Für die weitere Litteratur vergl. die zweite Note von Klein: Göttinger Nachrichten 19. Juli 1918.
In einer eben erschienenen Arbeit von Kneser (Math. Zeitschrift Bd. 2) handelt es sich um Aufstellung von Invarianten nach ähnlicher Methode.

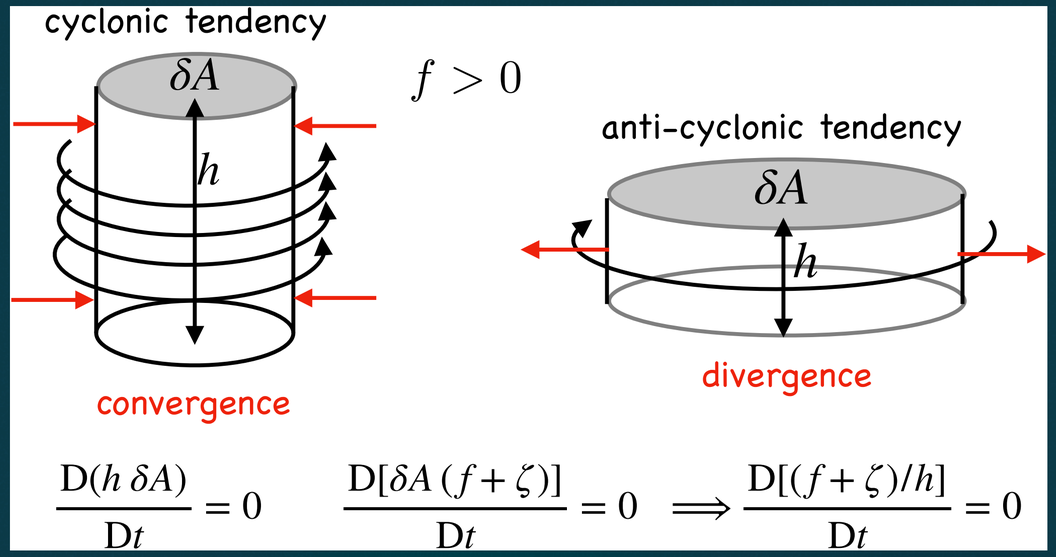
1

From the Princeton Institute for Advanced Studies website

Noether's theorem from 1918 states that for every continuous symmetry of a mechanical system, there exists a corresponding conserved physical quantity, and vice versa.

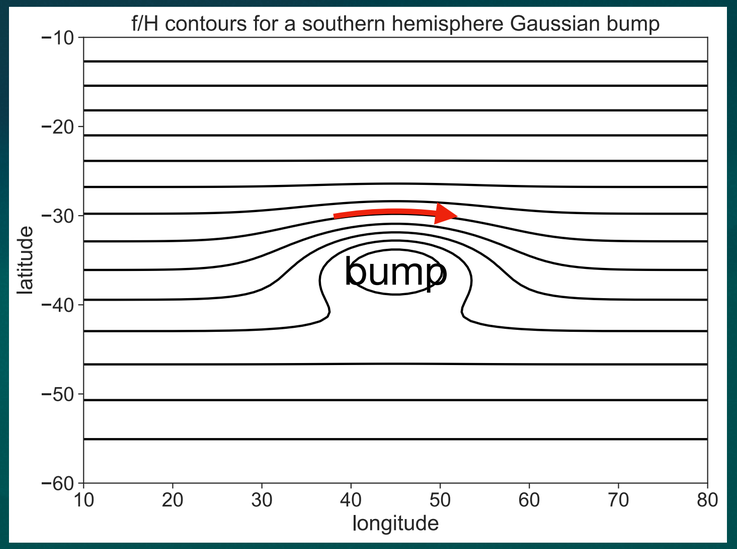
In geophysical fluid flows, such as in the ocean, it has its most nontrivial manifestation in potential vorticity conservation.

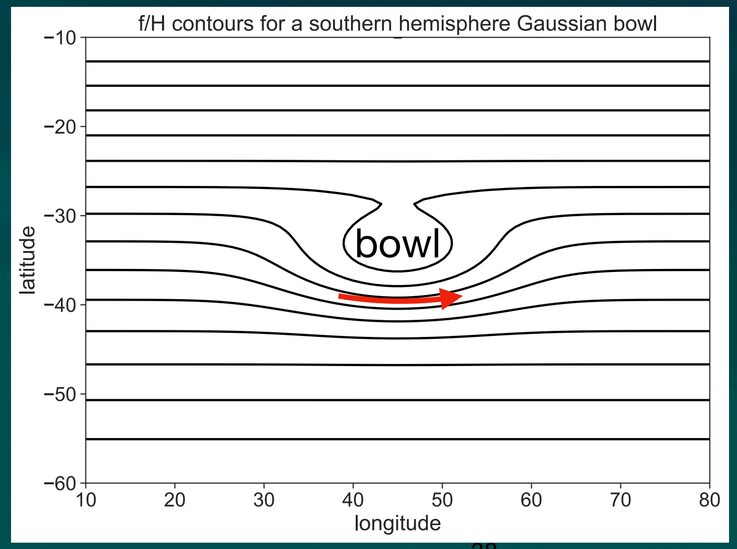
Illustrating Potential Vorticity Conservation



The conservation law for potential vorticity allows us to deduce motion without knowing the forces acting within the fluid.

This is a powerful application of Noether's theorem for theoretical ocean physics.





Closing Remarks

Understanding as Distinct from Knowledge

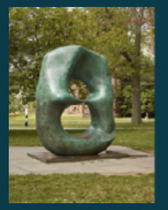
- Scientific understanding is characterized by grasping why or how a phenomenon occurs, going beyond the possession of knowledge, facts, or predictive accuracy.
- While scientific knowledge establishes the state of affairs, understanding recognizes the underlying structural, causal, or dynamical relations that govern phenomena.
- Understanding requires explanatory transparency: knowing which feedback loops, term balances, or physical constraints generate the output, enabling robust extrapolation beyond the domain where the data was originally sampled.

With input from Gemini AI

Scientific Elegance and Beauty

- W. Heisenberg: *Beauty is the proper conformity of the parts to one another and to the whole.*

- H. Moore: *A sculpture [*and theories*] should be viewed from all distances since new aspects are revealed at every scale.*



- Beautiful theories of ocean physics spring from a wonder and curiosity at the hows and whys of natural phenomena.
- Scientific description enhances, rather than diminishes, that wonder.
- Passion, creativity, and ideas deepen understanding. We need more to navigate an increasingly rough ride on this spinning Earth (AI is not sufficient).

LOCEAN Distinguished Ocean Scientist Colloquia

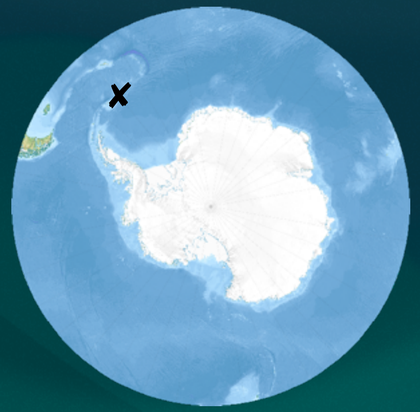
These speakers offer unique wisdom and experience as they tell their story about ocean physics, computer modeling, seagoing observations, and fundamental theory.

Some work primarily by going to sea; some work primarily with models and theory; and some make use of machine learning and AI as part of their toolbox.

You are encouraged to come to these lectures (in-person or virtually) and enjoy their energy and insights on facets of ocean and climate science!

Thanks for your attention!

Views from the Scotia Sea during April/May 2017 from the RRS JC Ross



Every ocean scientist should go to sea, to nurture the heart, mind, and soul.